

Network Security Groups and Application Security Groups

Introduction

This lab presents a scenario where one is responsible for designing and implementing secure virtual networking in Microsoft Azure using **Network Security Groups (NSGs)** and **Application Security Groups (ASGs)**. The organization's infrastructure includes two types of servers—Web Servers and Management Servers—each requiring specific network access controls. The objective is to ensure secure remote access to the management servers via RDP, while restricting such access to the web servers. Additionally, the web servers must be accessible via HTTP/HTTPS from the internet. Through this lab, one gains hands-on experience in deploying virtual networks, configuring security groups, applying inbound rules, and verifying that the configured network traffic filters function as intended—all within the East US Azure region.

Lab scenario

You have been asked to implement your organization's virtual networking infrastructure and test to ensure it is working correctly. In particular:

- The organization has two groups of servers: Web Servers and Management Servers.
- Each group of servers should be in its own Application Security Group.
- You should be able to RDP into the Management Servers, but not the Web Servers.
- The Web Servers should display the IIS web page when accessed from the internet.
- Network security group rules should be used to control network access.

Lab Objectives

The objective of this lab is to provide practical experience in building a secure virtual network environment in Azure using NSGs and ASGs. The lab emphasizes controlled access to resources through proper segmentation, rule-based filtering, and validation of network behavior based on the implemented configuration.

Tasks

By the end of this lab, I will have done the tasks below:

Exercise 1: Create the Virtual Networking Infrastructure

1. Created a virtual network with a single subnet.

2. Created two application security groups: one for Web Servers and another for Management Servers.
3. Created a network security group and associated it with the virtual network subnet.
4. Configured inbound NSG rules to allow:
 - HTTP and HTTPS access to the Web Servers.
 - RDP access to the Management Servers.

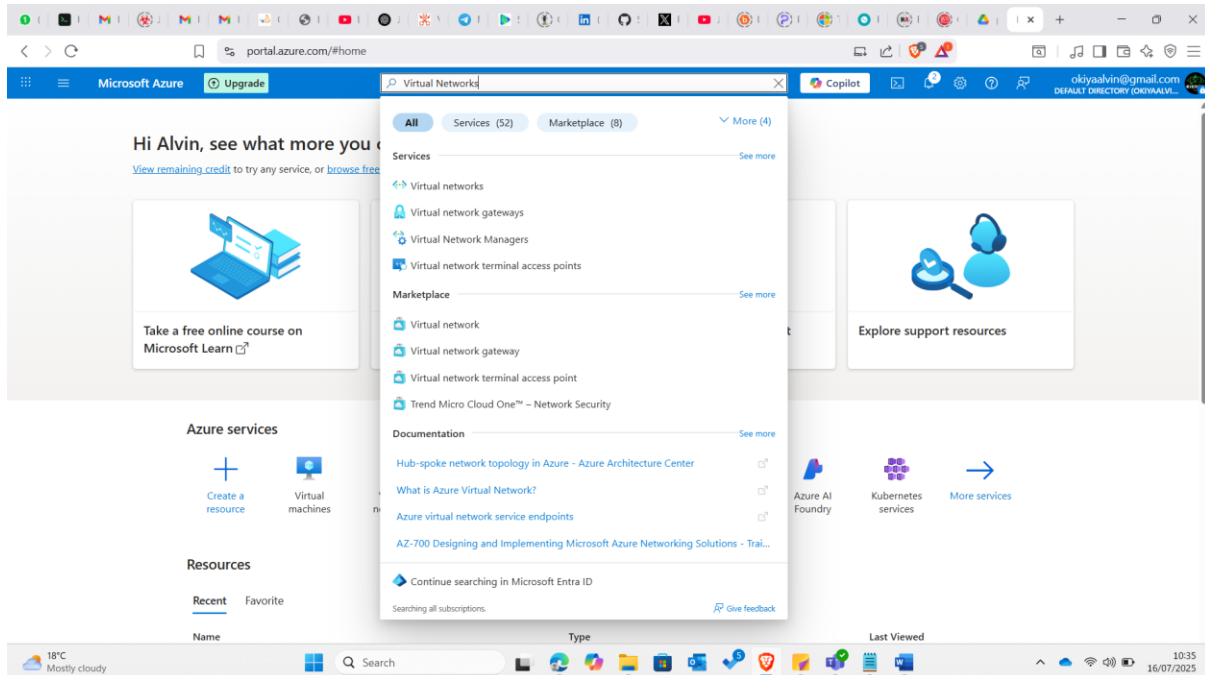
Exercise 2: Deploy Virtual Machines and Test Network Filters

1. Created a virtual machine to serve as a Web Server.
2. Created a virtual machine to serve as a Management Server.
3. Associated each virtual machine's network interface with the appropriate application security group.
4. Verified network access by:
 - Successfully connecting to the Management Server via RDP.
 - Successfully accessing the Web Server's IIS page from the internet.
 - Confirming that access restrictions were enforced as configured.

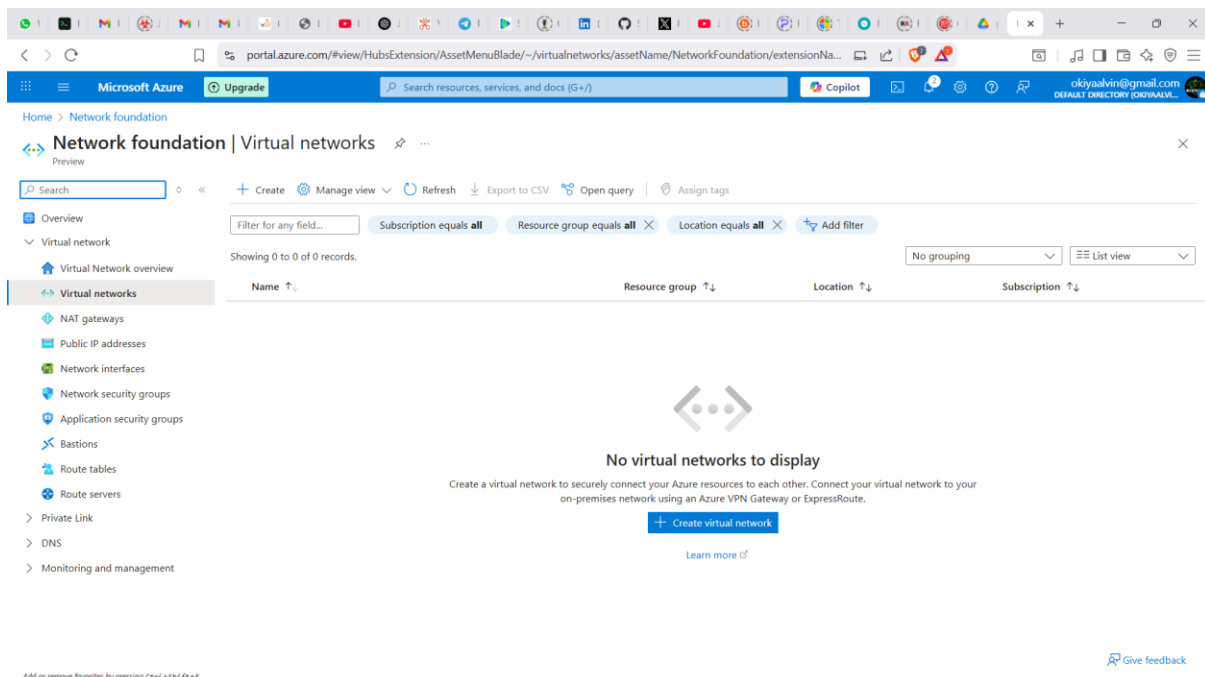
Exercise 1: Create the virtual networking infrastructure

Task 1: Creating a virtual network

To start this task I signed into my azure portal, navigated to the search bar, I searched for virtual networks:



After selecting the Virtual networks, I clicked on **create** on the Virtual Networks Blade



It took me to this section where I created a new resource group named **AZ500LAB07** and filled the other sections as shown:

The screenshot shows the 'Create virtual network' page in the Azure portal, specifically the 'Basics' tab. The page is titled 'Create virtual network' and includes a breadcrumb trail: 'Home > Network foundation | Virtual networks >'. Below the title, there are tabs for 'Basics', 'Security', 'IP addresses', 'Tags', and 'Review + create'. The 'Basics' tab is active, showing a brief description of virtual networks and a 'Learn more' link. The 'Project details' section contains two dropdown menus: 'Subscription' (set to 'Azure subscription 1') and 'Resource group' (set to '(New) AZ500LAB07'). The 'Instance details' section contains two dropdown menus: 'Virtual network name' (set to 'myVirtualNetwork') and 'Region' (set to '(US) East US'). A 'Deploy to an Azure Extended Zone' link is also present. The bottom of the page shows a Windows taskbar with the date and time as 10:42 on 16/07/2025.

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Home > Network foundation | Virtual networks >

Create virtual network

Basics Security IP addresses Tags Review + create

networks. vnet is similar to a traditional network that you operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * (New) AZ500LAB07 [Create new](#)

Instance details

Virtual network name * myVirtualNetwork

Region * (US) East US [Deploy to an Azure Extended Zone](#)

18°C Mostly cloudy 10:42 16/07/2025

I used the following settings in Ip address as they were the default settings and in line with the instructions:

The screenshot shows the 'Create virtual network' page in the Azure portal, specifically the 'IP addresses' tab. The page is titled 'Create virtual network' and includes a breadcrumb trail: 'Home > Network foundation | Virtual networks >'. Below the title, there are tabs for 'Basics', 'Security', 'IP addresses', 'Tags', and 'Review + create'. The 'IP addresses' tab is active, showing instructions on configuring the virtual network address space. The 'Add a subnet' section shows a table with the following data:

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-

The bottom of the page shows a 'Review + create' button and a 'Give feedback' link.

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Home > Network foundation | Virtual networks >

Create virtual network

Basics Security IP addresses Tags Review + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

+ Add a subnet

10.0.0.0/16 [Delete address space](#)

10.0.0.0 /16 65,536 addresses

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-

Add IPv4 address space

Previous Next Review + create Give feedback

I then clicked review and create, and after running validation, I clicked create.

The screenshot shows the 'Create virtual network' page in the Microsoft Azure portal, specifically the 'Review + create' tab. The page is divided into sections: Basics, Security, IP addresses, and Tags. The Basics section shows the subscription (Azure subscription 1), resource group (AZ500LAB07), name (myVirtualNetwork), and region (East US). The Security section shows that Azure Bastion, Azure Firewall, and Azure DDoS Network Protection are all disabled. The IP addresses section shows the address space (10.0.0.0/16) and subnet (default). The Tags section is empty. At the bottom, there are buttons for 'Previous', 'Next', and 'Create'. A 'Give feedback' link is also present.

Section	Item	Value
Basics	Subscription	Azure subscription 1
	Resource Group	AZ500LAB07
	Name	myVirtualNetwork
	Region	East US
Security	Azure Bastion	Disabled
	Azure Firewall	Disabled
	Azure DDoS Network Protection	Disabled
IP addresses	Address space	10.0.0.0/16 (65,536 addresses)
	Subnet	default (10.0.0.0/24) (256 addresses)
Tags		

And once the deployment was complete I proceeded to the next steps

The screenshot shows the 'myVirtualNetwork-1752652212071 | Overview' page in the Microsoft Azure portal. The page displays the deployment status as 'Your deployment is complete'. It shows the deployment name, subscription, resource group, start time, and correlation ID. There are links for 'Deployment details', 'Next steps', and 'Go to resource'. A 'Give feedback' link is also present. On the right side, there are sections for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

myVirtualNetwork-1752652212071 | Overview

Deployment

Search

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : myVirtualNetwork-1752652212071

Subscription : Azure subscription 1

Resource group : AZ500LAB07

Start time : 7/16/2025, 10:50:31 AM

Correlation ID : 3c50d8d7-484f-457c-afaa-0dd9e3d8d28d

Deployment details

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Cost management

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Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

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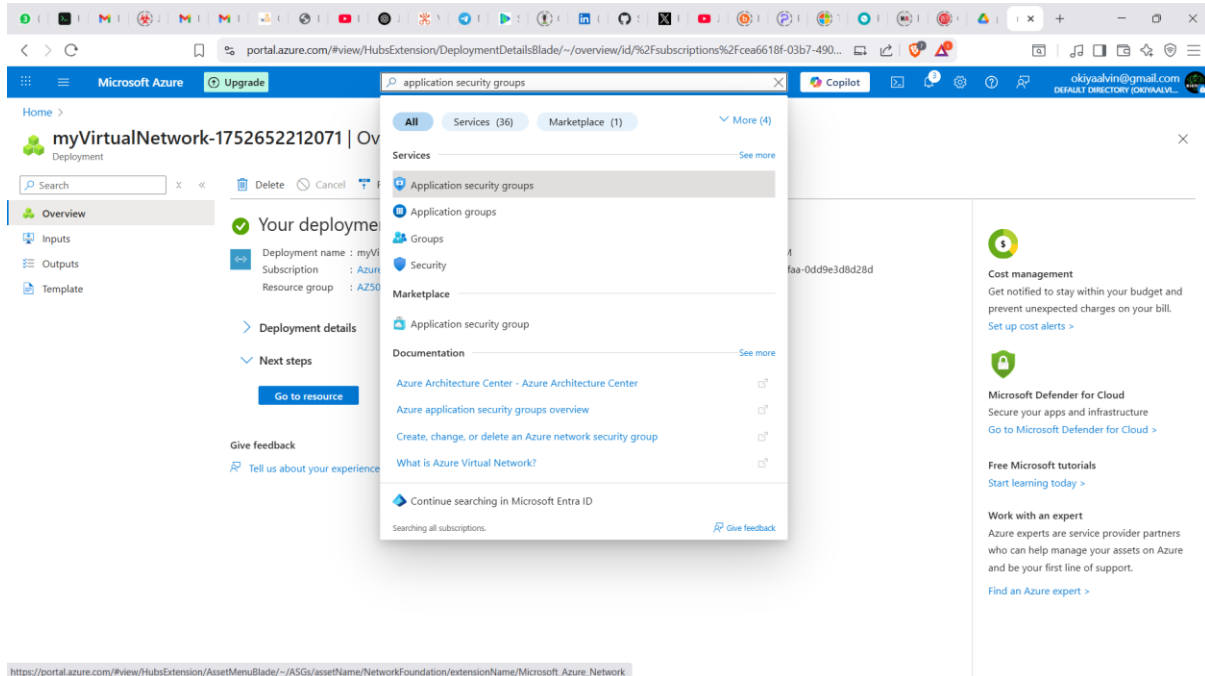
Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

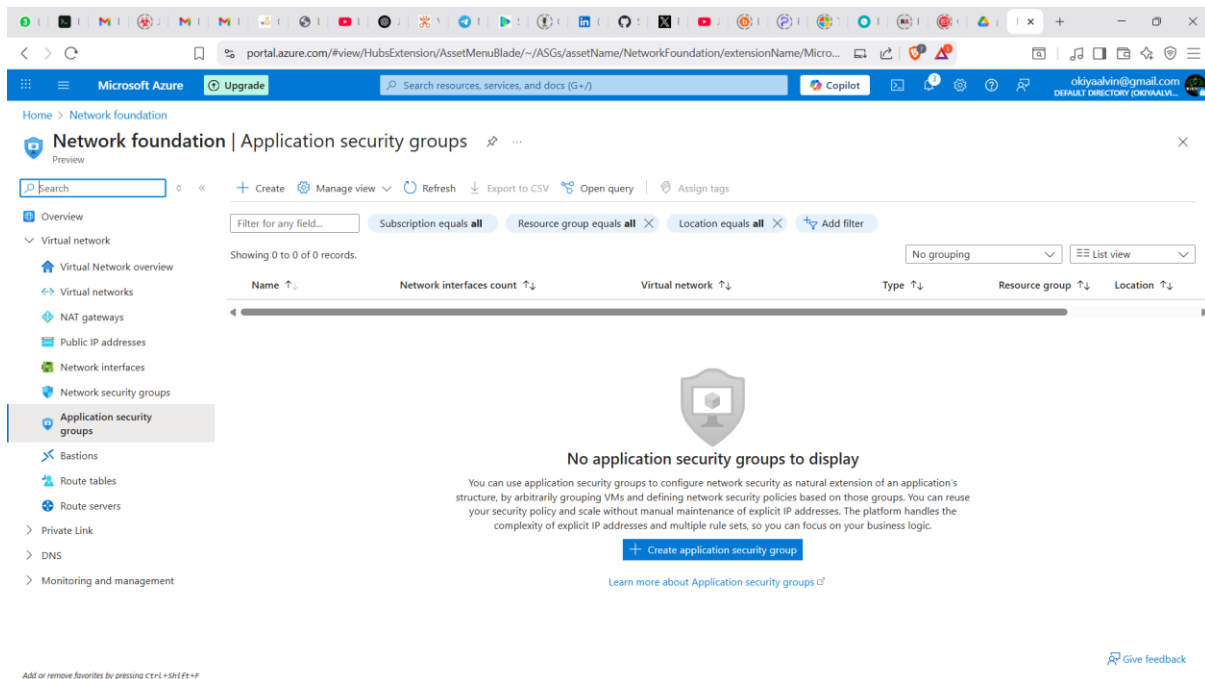
Find an Azure expert >

Task 2: Creating application security groups

In the azure portal at the search bar, I searched for **Application security groups** and pressed enter



Then on the applications security group, I clicked create



Then I filled the configurations as shown:

portal.azure.com/#create/Microsoft.ApplicationSecurityGroup

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Network foundation | Application security groups >

Create an application security group

Basics Tags Review + create

Project details

Subscription * Azure subscription 1

Resource group * AZ500LAB07
[Create new](#)

Instance details

Name * myAsgWebServer

Region * East US

[Review + create](#) < Previous Next: Tags > [Download a template for automation](#)

I then clicked “Review and create” then “create”

portal.azure.com/#create/Microsoft.ApplicationSecurityGroup

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Network foundation | Application security groups >

Create an application security group

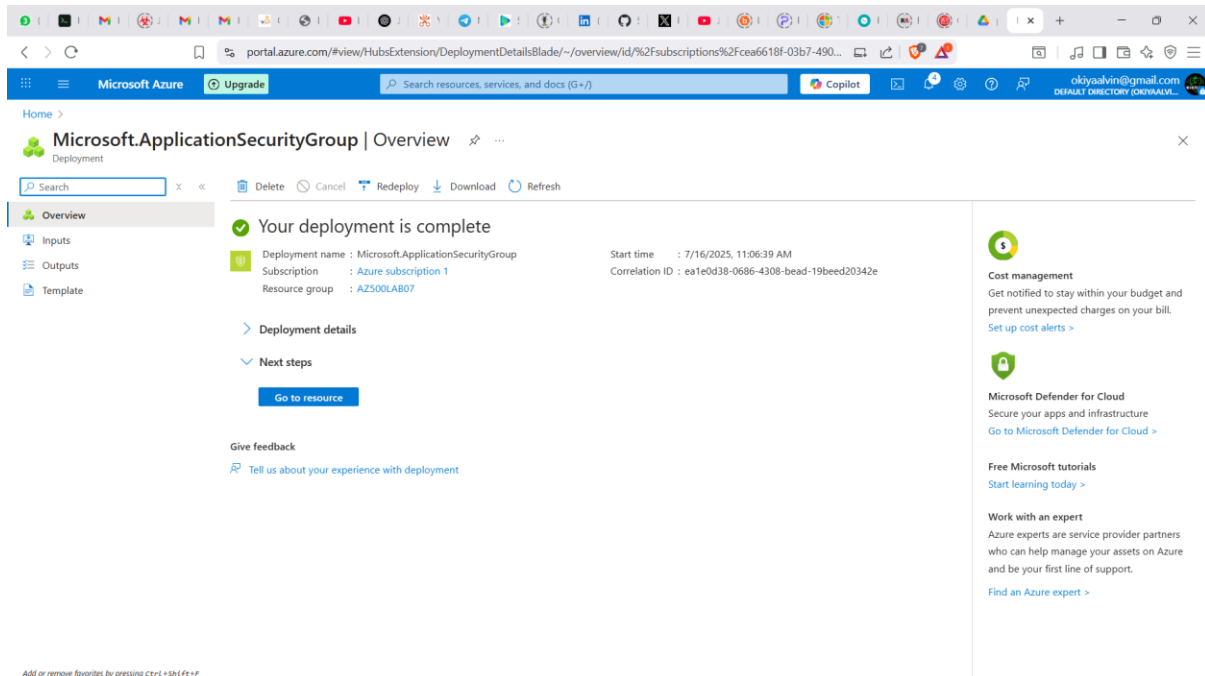
Basics Tags Review + create

Validation passed

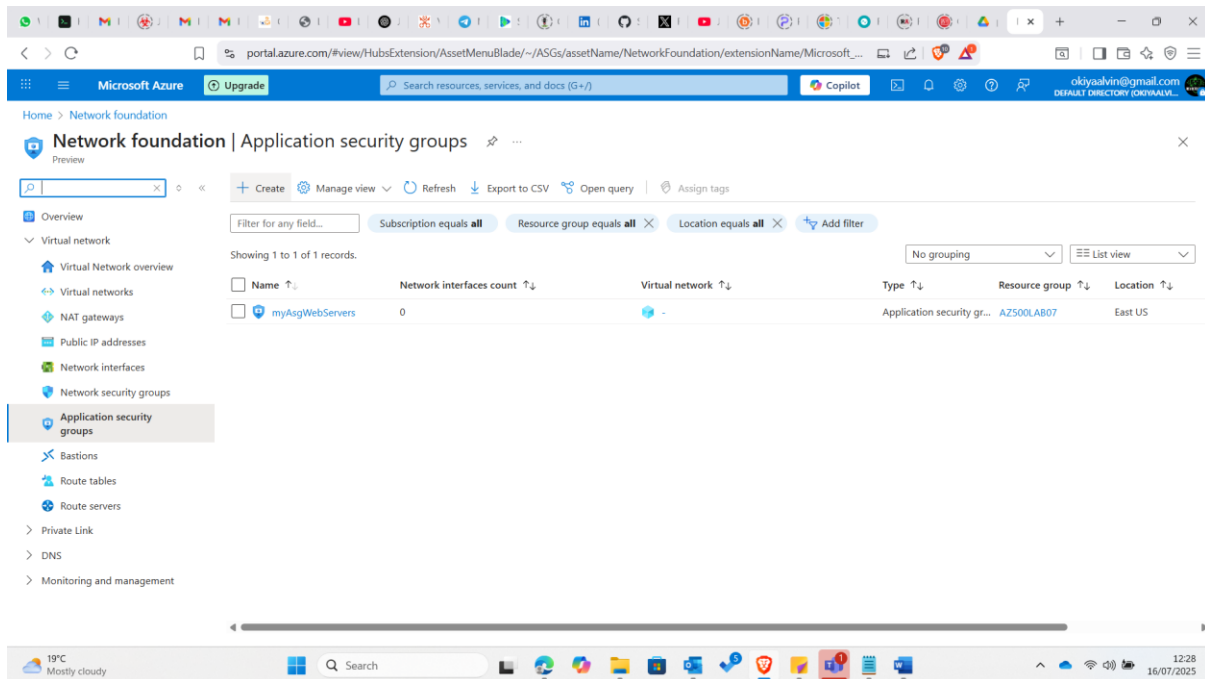
Summary

Subscription	Azure subscription 1
Resource group	AZ500LAB07
Location	East US
Name	myAsgWebServers

[Create](#) < Previous Next > [Download a template for automation](#)



I then navigated back to the **Application security groups** blade and clicked Create, in order to create a new Application security group for Management:



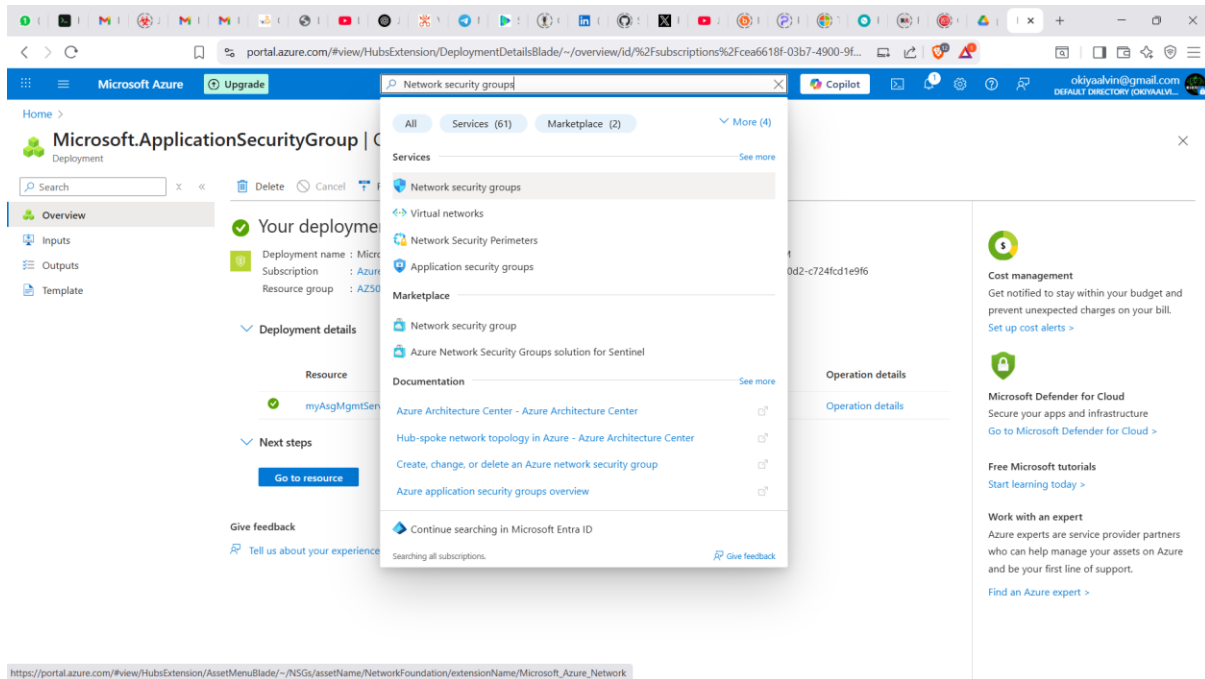
I then configured it with the settings below then clicked “review and create” then “create”:

The image shows two screenshots from the Microsoft Azure portal. The top screenshot is the 'Create an application security group' page. It has a breadcrumb trail: Home > Network foundation | Application security groups >. The page title is 'Create an application security group'. There are tabs for 'Basics', 'Tags', and 'Review + create'. Under 'Project details', 'Subscription' is 'Azure subscription 1' and 'Resource group' is 'AZ500LAB07'. Under 'Instance details', 'Name' is 'myAsgMgmtServers' and 'Region' is 'East US'. At the bottom, there are buttons for 'Review + create', '< Previous', 'Next: Tags >', and 'Download a template for automation'.

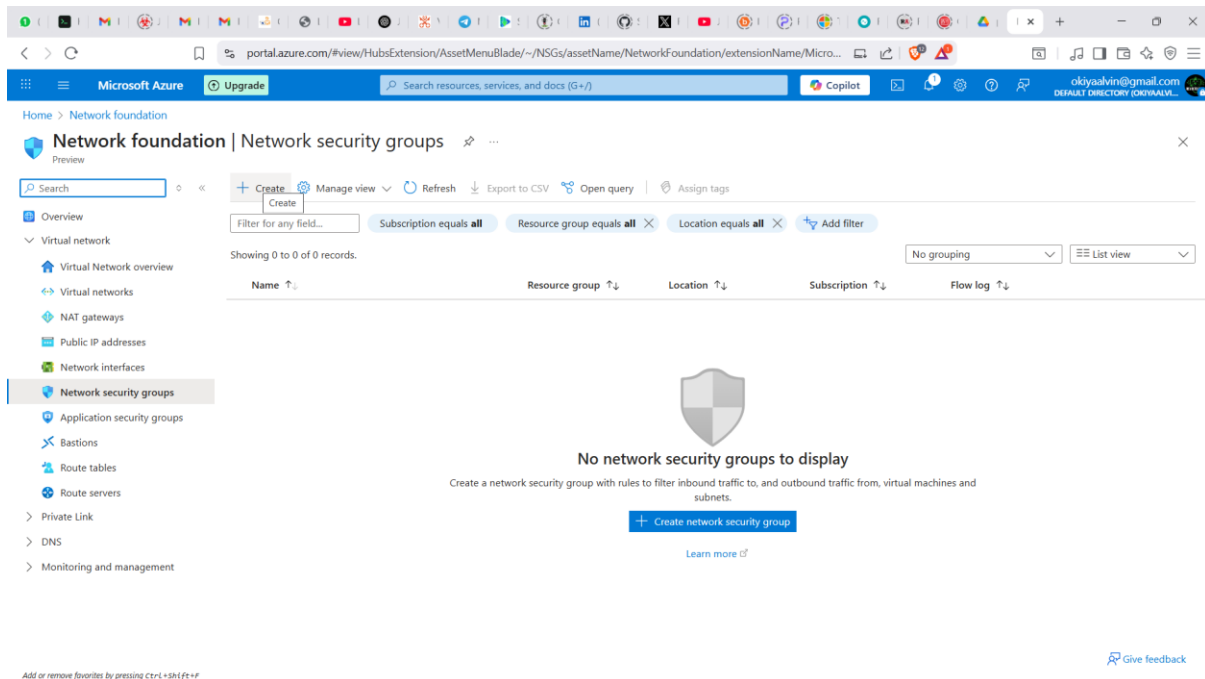
The bottom screenshot is the 'Overview' page for the 'Microsoft.ApplicationSecurityGroup' deployment. It shows a success message: 'Your deployment is complete'. Deployment details include: Deployment name: Microsoft.ApplicationSecurityGroup, Subscription: Azure subscription 1, Resource group: AZ500LAB07, Start time: 7/16/2025, 12:32:45 PM, and Correlation ID: 99ef7280-2ce4-4d76-a0d2-c724fd1e9f6. A table shows the deployment details for the resource 'myAsgMgmtServers', which is an 'Application security group' with a status of 'OK'. The 'Next steps' section has a 'Go to resource' button. The right sidebar contains links for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

Task 3: Create a network security group and associate the NSG to the subnet

In this step I navigated to the search button in the azure portal and searched for **Network security groups** as shown:



I then clicked create:



Then configured it as shown:

portal.azure.com/#create/Microsoft.NetworkSecurityGroup

Create network security group

Validation passed

Basics Tags **Review + create**

Basics

Subscription: Azure subscription 1
Resource group: AZ500LAB07
Region: East US
name: myNsg

Tags

None

Create < Previous Next > Download a template for automation

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fcea6618f-03b7-490...

Microsoft.NetworkSecurityGroup-20250716123714 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs
Outputs
Template

Your deployment is complete

Deployment name: Microsoft.NetworkSecurityGroup-20250716123714 Start time: 7/16/2025, 12:39:29 PM
Subscription: Azure subscription 1 Correlation ID: 04c720d4-3c17-4e62-bd81-fa6691c672c2
Resource group: AZ500LAB07

Deployment details

Resource	Type	Status	Operation details
myNsg	Network security group	OK	Operation details

Next steps

[Go to resource](#)

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Microsoft Defender for Cloud

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Free Microsoft tutorials

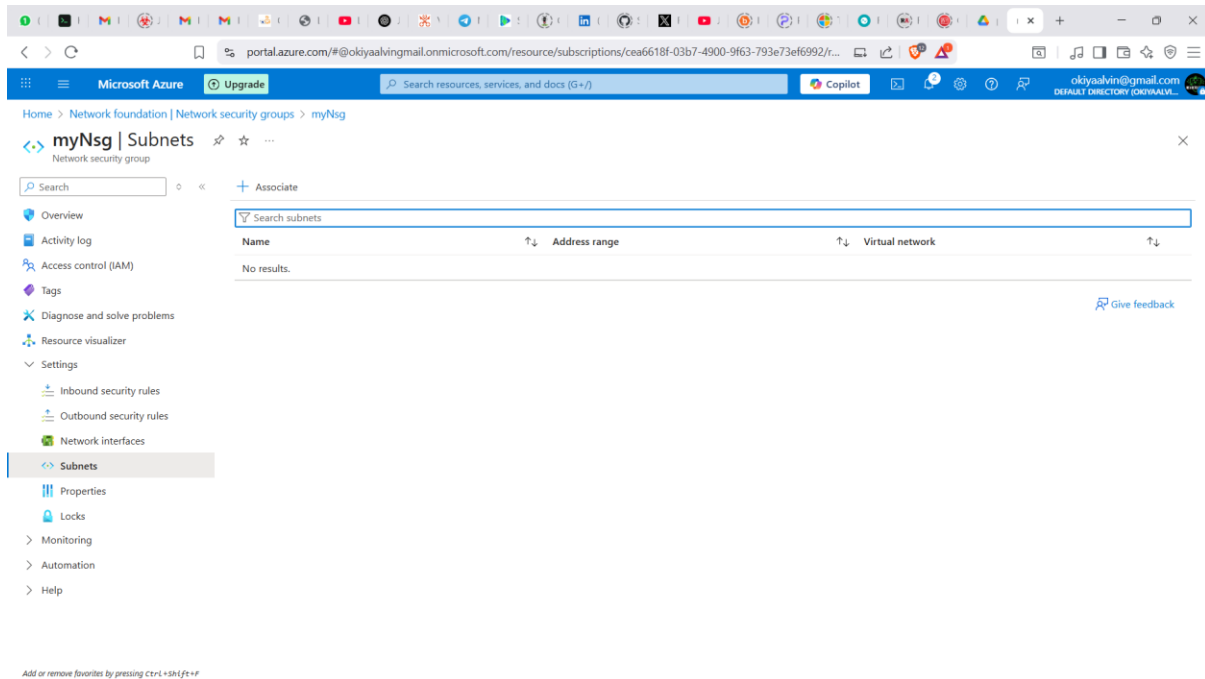
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Work with an expert

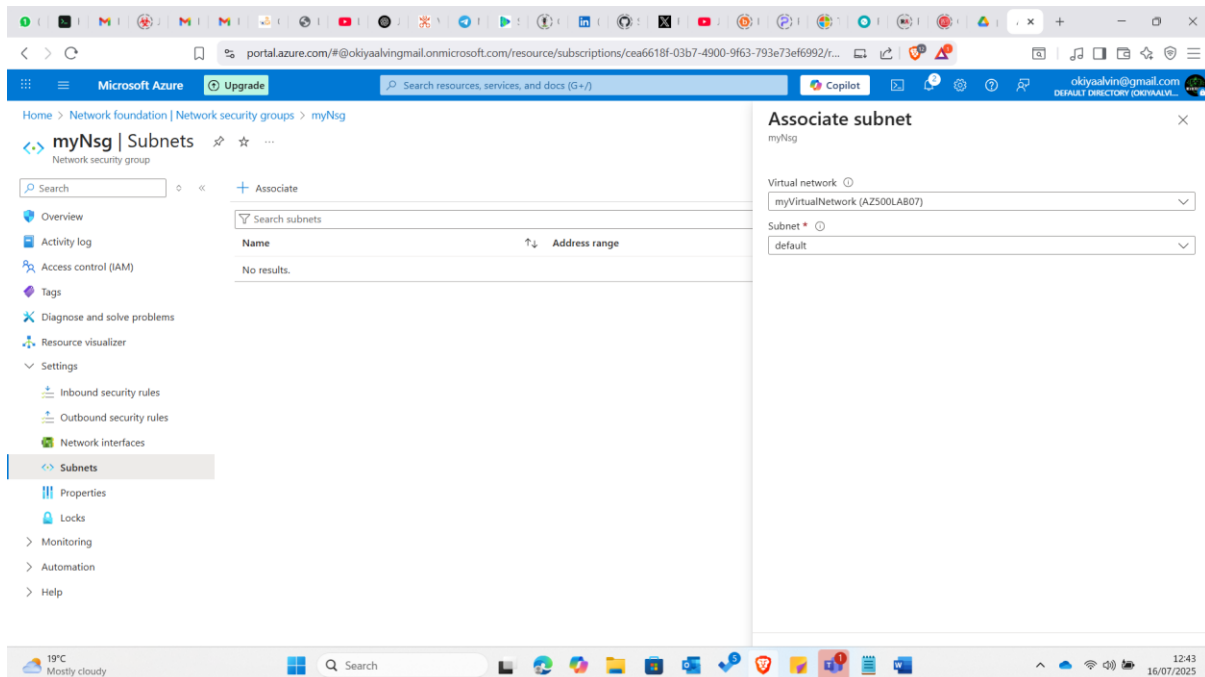
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In the Azure portal, I navigated back to the **Network security groups** blade and clicked the **myNsg** entry and in the settings section I clicked on subnets, then +Associate

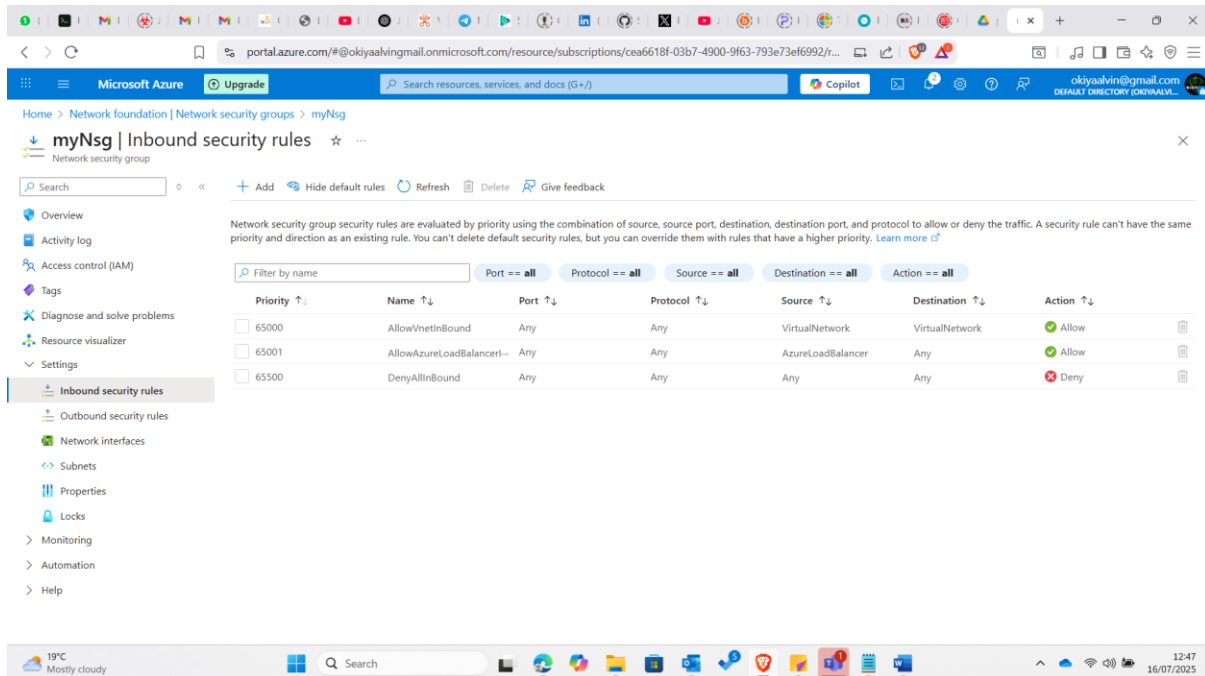


On the associate subnets blade I specified the following:



Task 4: Create inbound NSG security rules to all traffic to web servers and RDP to the servers

On the **myNsg** blade, in the **Settings** section, I clicked **Inbound security rules** and reviewed the existing rules:



Then clicked on “+ Add” and on the **inbound security rule** blade, I specified the following settings to allow TCP ports 80 and 443 to the **myAsgWebServers** application security group:

Setting	Value
Destination	in the drop-down list, select Application security group and then click myAsgWebServers
Destination port ranges	80,443
Protocol	TCP
Priority	100
Name	Allow-Web-All

Microsoft Azure portal showing the configuration of the 'myNsg | Inbound security rules' page. The 'Allow-Web-All' rule is selected, and its configuration is shown in the right-hand pane.

myNsg | Inbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and direction as an existing rule. You can't delete default security rules, but you can override them with rules.

Priority	Name	Port	Protocol
100	Allow-Web-All	8080	TCP
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalancerInBound	Any	Any
65500	DenyAllInBound	Any	Any

Allow-Web-All

Destination application security groups: myAsgWebServers

Service: Custom

Destination port ranges: 80,443

Protocol: TCP

Action: Allow

Priority: 100

Then I clicked on Add, and it successfully created the inbound security group for the web servers.

Next was to create an inbound security group for the Management server.

On the **myNsg** blade, in the **Settings** section, I clicked **Inbound security rules** > + Add.

Microsoft Azure portal showing the configuration of the 'Add inbound security rule' page. The configuration is shown in the right-hand pane.

Add inbound security rule

Destination application security groups: myAsgMgmtServers

Service: Custom

Destination port ranges: 3389

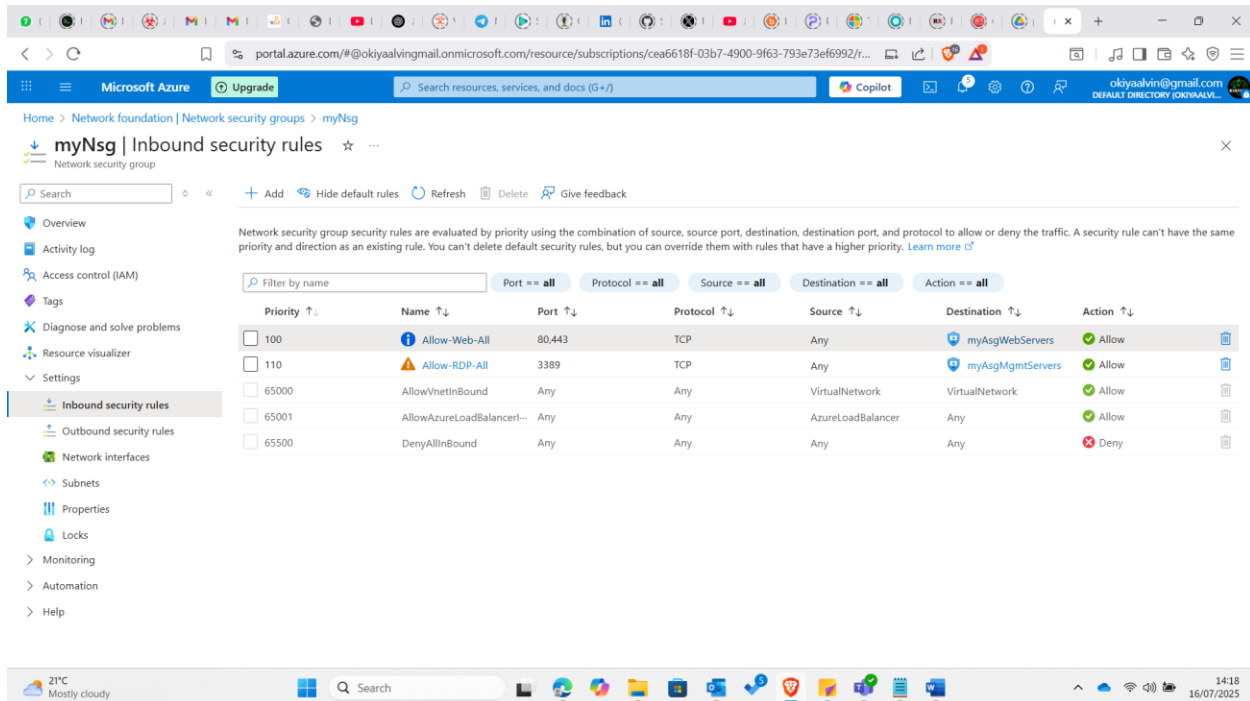
Protocol: TCP

Action: Allow

Priority: 110

Name: Allow-RDP-All

Configured the settings as shown and clicked on add.

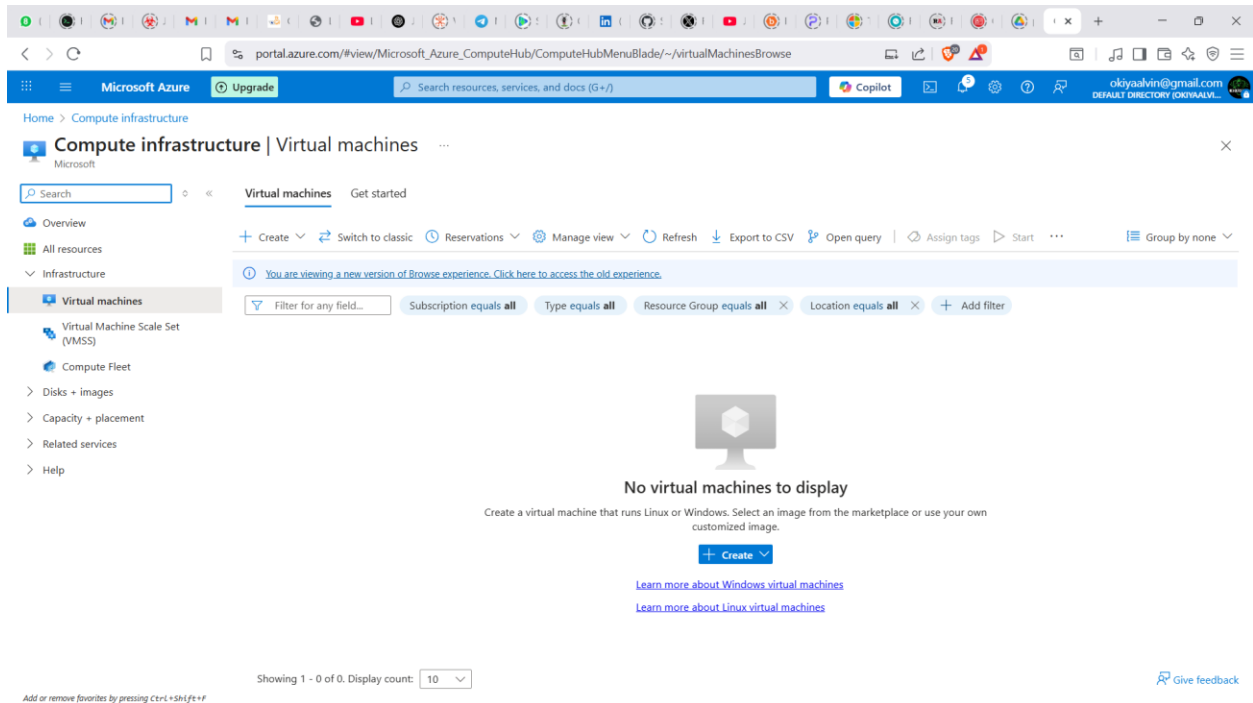


Exercise 2: Deploy virtual machines and test network filters

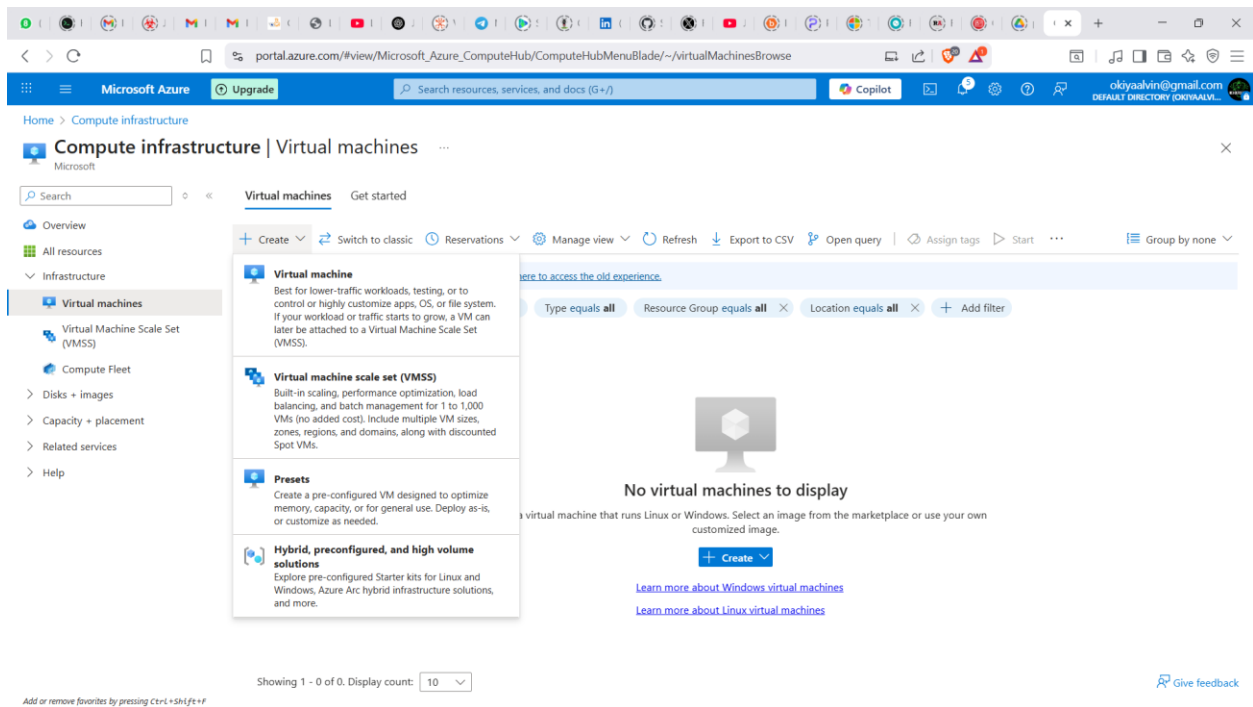
Task 1: Create a virtual machine to use as a web server.

In this task, I was supposed to create a virtual machine to use as a web server.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Virtual machines** and selected the virtual machine option, then clicked on Create:



Then on the drop down I selected Azure Virtual machine:



Upon clicking create I filled in the configuration details with the following details:

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ500LAB07
Virtual machine name	myVmWeb
Region	(US)East US
Availability options	No infrastructure redundancy required
Security type	Standard
Image	Windows Server 2022 Datacenter: Azure Edition- x64 Gen2
Size	Standard D2s v3
Username	Student
Password	Please create your own password and record it for future reference in subsequent labs
Confirm password	Retype your password
Public inbound ports	None
Would you like to use an existing Windows Server License	No

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure Upgrade Search resources, services, and docs (G+/J) Copilot okyaalvin@gmail.com

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1
Resource group * AZ500LAB07
[Create new](#)

Instance details

Virtual machine name * myVmWeb ✓
Region * (US) East US ✓
Availability options No infrastructure redundancy required ✓
Security type Standard ✓
Image * Windows Server 2022 Datacenter: Azure Edition - x64 Gen2 (free services el) ✓
[See all images](#) | [Configure VM generation](#)
This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)
VM architecture ☐ Arm64 ☒ x64

< Previous Next: Disks > Review + create Give feedback

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

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Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Size * Standard_D2s_v3 - 2 vcpus, 8 GiB memory (\$137.24/month) ✓
[See all sizes](#)

Enable Hibernation ☐
Hibernation is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#)

Administrator account

Username * Student ✓
Password * ✓
Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☒ None ☐ Allow selected ports

< Previous Next: Disks > Review + create Give feedback

Next I clicked on Next: Disks and on the Disks blade I selected the Disk type to **Standard HDD** then clicked on Next: Networking:

The screenshot shows the 'Create a virtual machine' blade in the Microsoft Azure portal. The 'Disks' tab is selected, showing configuration options for the OS disk. The OS disk size is set to 'Image default (127 GiB)' and the OS disk type is 'Standard HDD (locally-redundant storage)'. There is a note stating: 'The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.' The 'Delete with VM' checkbox is checked, and the 'Key management' is set to 'Platform-managed key'. The 'Enable Ultra Disk compatibility' checkbox is unchecked, with a note: 'Ultra disk is supported in Availability Zone(s) 1,2,3 for the selected VM size Standard_D2s_v3.' Below these options is a table for 'Data disks for myVmWeb' with columns: LUN, Name, Size (GiB), Disk type, Host caching, and Delete with VM. At the bottom, there are navigation buttons: '< Previous', 'Next: Networking >', and 'Review + create'. A 'Give feedback' link is also present.

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Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size Image default (127 GiB)

OS disk type * Standard HDD (locally-redundant storage)
The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ☒

Key management Platform-managed key

Enable Ultra Disk compatibility ☐
Ultra disk is supported in Availability Zone(s) 1,2,3 for the selected VM size Standard_D2s_v3.

Data disks for myVmWeb

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
-----	------	------------	-----------	--------------	----------------

[Create and attach a new disk](#) [Attach an existing disk](#)

< Previous Next: Networking > Review + create [Give feedback](#)

On the **Networking** tab of the **Create a virtual machine** blade, I selected the previously created network **myVirtualNetwork**, and Under **NIC network security group** I selected **None**:

The screenshot shows the 'Create a virtual machine' blade in the Microsoft Azure portal, with the 'Networking' tab selected. It shows the configuration for the network interface. The 'Virtual network' is set to 'myVirtualNetwork' with a 'Create new' link. The 'Subnet' is set to 'default (10.0.0.0/24)' with a 'Manage subnet configuration' link. The 'Public IP' is set to '[new] myVmWeb-ip' with a 'Create new' link. The 'NIC network security group' is set to 'None' (radio button selected), with options for 'Basic' and 'Advanced'. A warning message states: 'All ports on this virtual machine may be exposed to the public internet. This is a security risk. Use a network security group to limit public access to specific ports. You can also select a subnet that already has network security groups defined or remove the public IP address.' At the bottom, there are navigation buttons: '< Previous', 'Next: Management >', and 'Review + create'. A 'Give feedback' link is also present.

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Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Basics Disks **Networking** Management Monitoring Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * myVirtualNetwork
[Create new](#)

Subnet * default (10.0.0.0/24)
[Manage subnet configuration](#)

Public IP [new] myVmWeb-ip
[Create new](#)

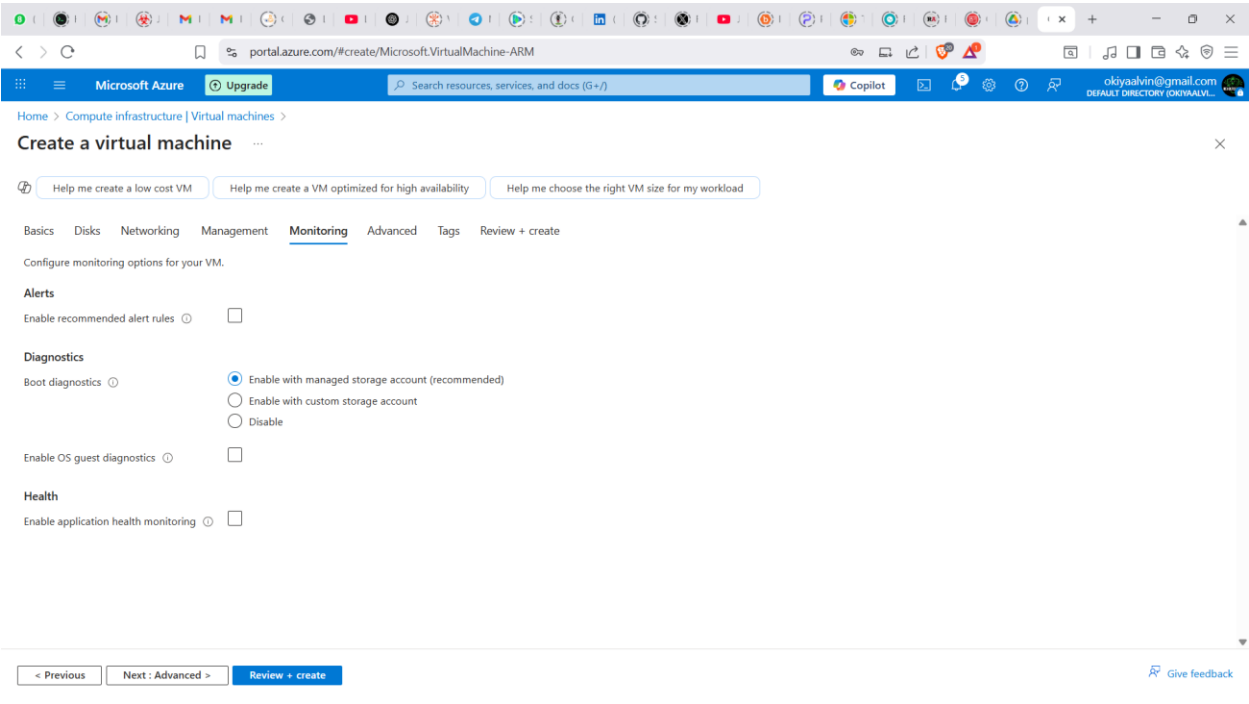
NIC network security group ☒ None ☐ Basic ☐ Advanced

All ports on this virtual machine may be exposed to the public internet. This is a security risk. Use a network security group to limit public access to specific ports. You can also select a subnet that already has network security groups defined or remove the public IP address.

< Previous Next: Management > Review + create [Give feedback](#)

Then I Clicked **Next: Management >**, then **Next: Monitoring >**. On the Monitoring tab of the **Create a virtual machine** blade, I verified the following setting:

Setting	Value
Boot diagnostics	Enabled with managed storage account (recommended)



Then I clicked **review and create** and after a successful verification, I clicked **create**.

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fcea6618f-03b...

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Home > CreateVm-MicrosoftWindowsServer.WindowsServer-202-20250716142624 | Overview

Deployment

Search x « Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Your deployment is complete

Deployment name: CreateVm-MicrosoftWindowsServer.WindowsSe... Start time: 7/16/2025, 2:49:40 PM
Subscription: Azure subscription 1 Correlation ID: c295bdef-3d13-4693-a4cb-66eb7f509c59

Resource group: AZ500LAB07

Deployment details

Resource	Type	Status	Operation details
myVmWeb	Microsoft.Compute/virtualMachines	OK	Operation details
myVmWeb317	Microsoft.Network/networkInterfa...	OK	Operation details
myVmWeb-ip	Microsoft.Network/publicAddr...	OK	Operation details

Next steps

- Setup auto-shutdown Recommended
- Monitor VM health, performance and network dependencies Recommended
- Run a script inside the virtual machine Recommended

[Go to resource](#) [Create another VM](#)

Give feedback
[Tell us about your experience with deployment](#)

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[Set up cost alerts >](#)

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Task 2: Create a virtual machine to use as a management server.

In the Azure portal, I navigated back to the **Virtual machines** blade, clicked **+ Create**, and, in the dropdown list, then clicked **+ Azure virtual machine**

portal.azure.com/#view/Microsoft_Azure_ComputeHub/ComputeHubMenuBlade/~/virtualMachinesBrowse

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Home > Compute infrastructure

Compute infrastructure | Virtual machines

Microsoft

Search « Virtual machines Get started

Overview

- All resources
- Infrastructure
- Virtual machines**
 - Virtual Machine Scale Set (VMSS)
 - Compute Fleet
- Disks + images
- Capacity + placement
- Related services
- Help

+ Create Switch to classic Reservations Manage view Refresh Export to CSV Open query Assign tags Start Group by none

Virtual machine
Best for lower-traffic workloads, testing, or to control or highly customize apps, OS, or file system. If your workload or traffic starts to grow, a VM can later be attached to a Virtual Machine Scale Set (VMSS).

Virtual machine scale set (VMSS)
Built-in scaling, performance optimization, load balancing, and batch management for 1 to 1,000 VMs (no added cost). Include multiple VM sizes, zones, regions, and domains, along with discounted Spot VMs.

Presets
Create a pre-configured VM designed to optimize memory, capacity, or for general use. Deploy as-is, or customize as needed.

Hybrid, preconfigured, and high volume solutions
Explore pre-configured Starter kits for Linux and Windows, Azure Arc hybrid infrastructure solutions, and more.

[here to access the old experience.](#)

Type equals all Resource Group equals all Location equals all Add filter

Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
ipt...	AZ500LAB07	East US	Running	Windows	Standard_D2s_v3	172.203.139.180 1

Showing 1 - 1 of 1. Display count: 10 Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

I then configured as shown:

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal, specifically the 'Basics' tab. The page is titled 'Create a virtual machine' and includes a breadcrumb trail: 'Home > Compute infrastructure | Virtual machines >'. Below the title, there are three help links: 'Help me create a low cost VM', 'Help me create a VM optimized for high availability', and 'Help me choose the right VM size for my workload'. The 'Basics' tab is selected, and the page content includes a description of creating a VM, a warning about subscription eligibility, and a 'Project details' section. The 'Project details' section contains two dropdown menus: 'Subscription' (set to 'Azure subscription 1') and 'Resource group' (set to 'AZ500LAB07'). Below this is the 'Instance details' section, which includes three dropdown menus: 'Virtual machine name' (set to 'myVMMgmt'), 'Region' (set to '(US) East US'), and 'Availability options' (set to 'No infrastructure redundancy required'). At the bottom of the page, there are navigation buttons: '< Previous', 'Next: Disks >', and 'Review + create'. A 'Give feedback' link is also present in the bottom right corner.

Microsoft Azure Upgrade Search resources, services, and docs (G+J) Copilot okyaalvin@gmail.com DEFAULT DIRECTORY (OKYAALV...)

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * AZ500LAB07 [Create new](#)

Instance details

Virtual machine name * myVMMgmt

Region * (US) East US

Availability options No infrastructure redundancy required

< Previous Next: Disks > Review + create [Give feedback](#)

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal, specifically the 'Disks' tab. The page is titled 'Create a virtual machine' and includes a breadcrumb trail: 'Home > Compute infrastructure | Virtual machines >'. Below the title, there are three help links: 'Help me create a low cost VM', 'Help me create a VM optimized for high availability', and 'Help me choose the right VM size for my workload'. The 'Disks' tab is selected, and the page content includes a description of creating a VM, a warning about subscription eligibility, and a 'Project details' section. The 'Project details' section contains two dropdown menus: 'Subscription' (set to 'Azure subscription 1') and 'Resource group' (set to 'AZ500LAB07'). Below this is the 'Instance details' section, which includes three dropdown menus: 'Virtual machine name' (set to 'myVMMgmt'), 'Region' (set to '(US) East US'), and 'Availability options' (set to 'No infrastructure redundancy required'). At the bottom of the page, there are navigation buttons: '< Previous', 'Next: Disks >', and 'Review + create'. A 'Give feedback' link is also present in the bottom right corner.

Microsoft Azure Upgrade Search resources, services, and docs (G+J) Copilot okyaalvin@gmail.com DEFAULT DIRECTORY (OKYAALV...)

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Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Disks Basics Networking Management Monitoring Advanced Tags Review + create

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Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * AZ500LAB07 [Create new](#)

Instance details

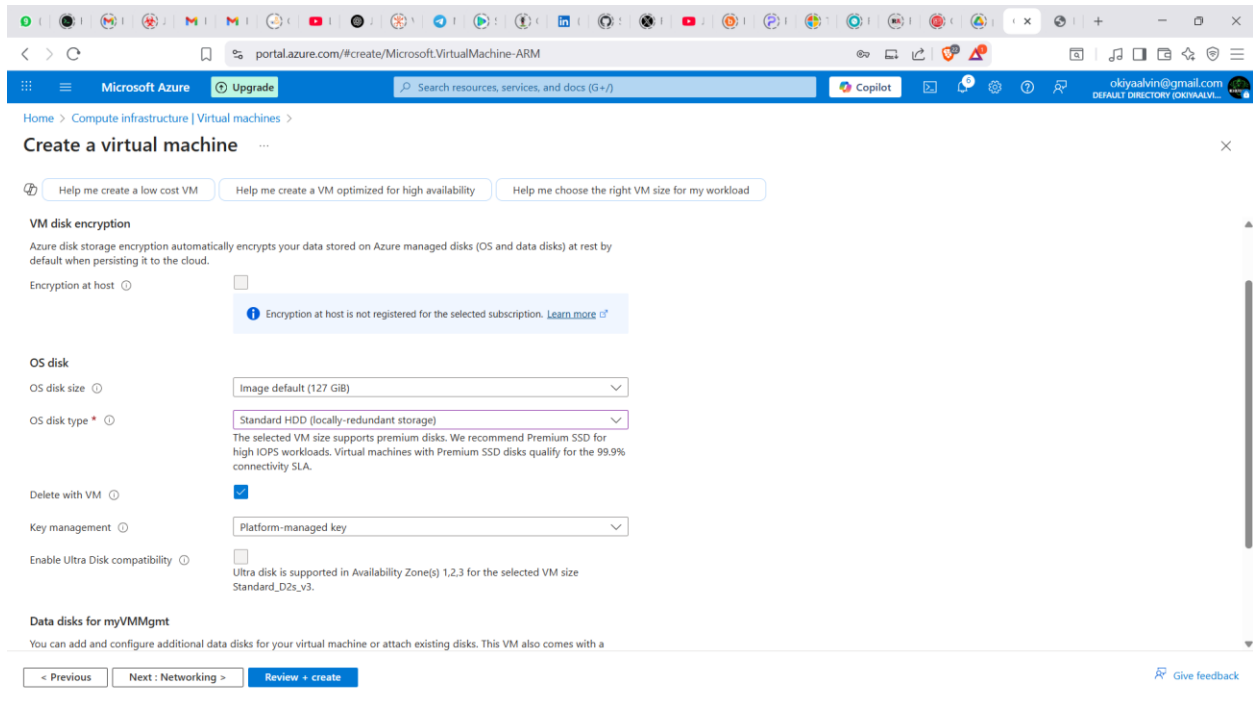
Virtual machine name * myVMMgmt

Region * (US) East US

Availability options No infrastructure redundancy required

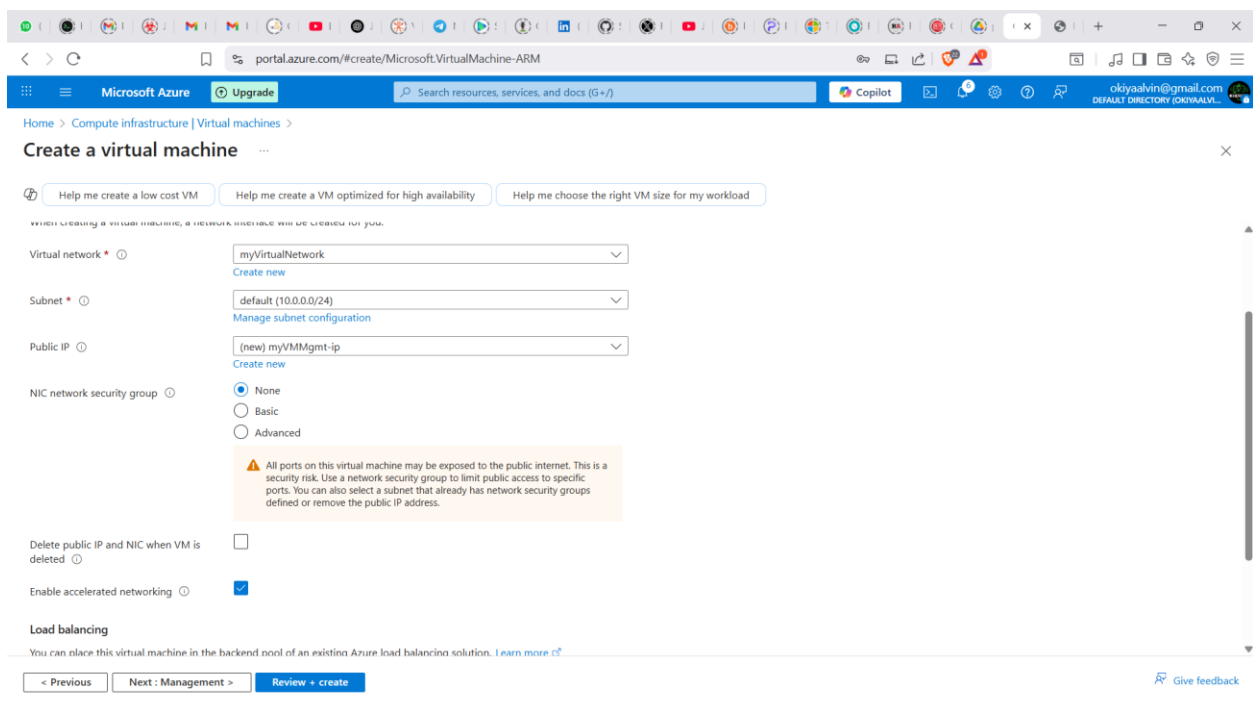
< Previous Next: Disks > Review + create [Give feedback](#)

Then I clicked on **Next: Disks >** and, on the **Disks** tab of the **Create a virtual machine** blade, I set the **OS disk type** to **Standard HDD** and clicked **Next: Networking >**



The screenshot shows the 'Create a virtual machine' blade in the Azure portal, specifically the 'Disks' tab. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there are three tabs: 'Help me create a low cost VM', 'Help me create a VM optimized for high availability', and 'Help me choose the right VM size for my workload'. The 'Disks' tab is active, showing the 'VM disk encryption' section with a checkbox for 'Encryption at host' and a message indicating it's not registered for the selected subscription. Below this is the 'OS disk' section with a dropdown for 'OS disk size' (Image default (127 GiB)) and a dropdown for 'OS disk type' (Standard HDD (locally-redundant storage)). There is also a checkbox for 'Delete with VM' and a dropdown for 'Key management' (Platform-managed key). The 'Enable Ultra Disk compatibility' section has a checkbox and a message about availability zones. At the bottom, there is a 'Data disks for myVMMgmt' section with a message about adding and configuring additional data disks. Navigation buttons at the bottom include '< Previous', 'Next: Networking >', and 'Review + create'.

Then proceeded to the Networking section where I confirmed that the network selected was **myVirtualNetwork** and Under **NIC network security group** I selected **None**.



The screenshot shows the 'Create a virtual machine' blade in the Azure portal, specifically the 'Networking' tab. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there are three tabs: 'Help me create a low cost VM', 'Help me create a VM optimized for high availability', and 'Help me choose the right VM size for my workload'. The 'Networking' tab is active, showing the 'Virtual network' section with a dropdown for 'Virtual network' (myVirtualNetwork) and a link to 'Create new'. Below this is the 'Subnet' section with a dropdown for 'Subnet' (default (10.0.0.0/24)) and a link to 'Manage subnet configuration'. The 'Public IP' section has a dropdown for 'Public IP' ((new) myVMMgmt-ip) and a link to 'Create new'. The 'NIC network security group' section has three radio buttons: 'None' (selected), 'Basic', and 'Advanced'. There is a warning message about public internet exposure. Below this is the 'Delete public IP and NIC when VM is deleted' section with a checkbox. The 'Enable accelerated networking' section has a checkbox. At the bottom, there is a 'Load balancing' section with a message about placing the virtual machine in the backend pool of an existing Azure load balancing solution. Navigation buttons at the bottom include '< Previous', 'Next: Management >', and 'Review + create'.

Next I proceeded to click **Next: Management >**, then clicked **Next: Monitoring >**. On the **Monitoring** tab of the **Create a virtual machine** blade, I verified the following setting:

The screenshot shows the 'Create a virtual machine' blade in the Microsoft Azure portal. The 'Monitoring' tab is selected, and the 'Alerts' section is expanded. The 'Enable recommended alert rules' checkbox is checked. The 'Diagnostics' section is also expanded, showing 'Boot diagnostics' and 'Enable OS guest diagnostics'. The 'Boot diagnostics' section has three options: 'Enable with managed storage account (recommended)' (selected), 'Enable with custom storage account', and 'Disable'. The 'Enable OS guest diagnostics' checkbox is checked. The 'Health' section has one option: 'Enable application health monitoring' (unchecked). At the bottom, there are three buttons: '< Previous', 'Next: Advanced >', and 'Review + create'.

Then clicked on “review + create” and after successful validation, I clicked create:

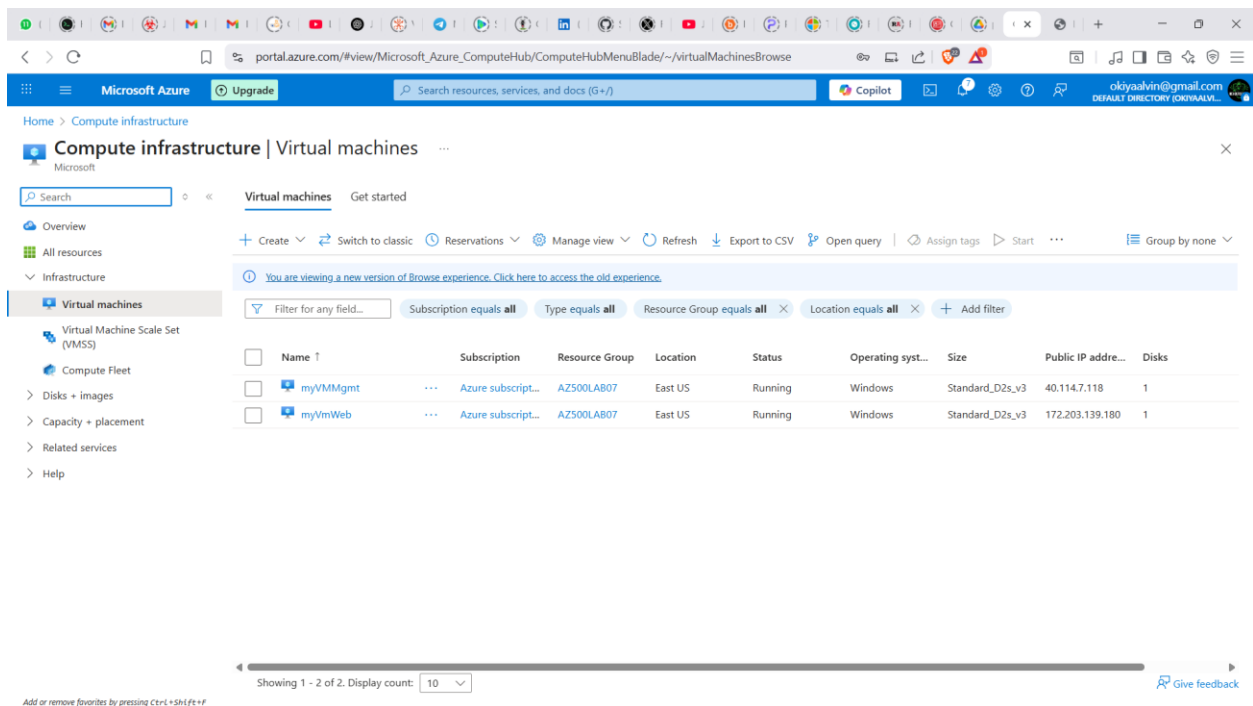
The screenshot shows the 'CreateVm-MicrosoftWindowsServer.WindowsServer-202-20250716145528 | Overview' blade in the Microsoft Azure portal. The 'Overview' tab is selected, and the 'Your deployment is complete' message is displayed. The deployment details table shows three resources: 'myVMMgmt', 'myvmgmt709', and 'myVMMgmt-ip', all with a status of 'OK'. The 'Next steps' section includes links for 'Setup auto-shutdown', 'Monitor VM health, performance and network dependencies', and 'Run a script inside the virtual machine'. The 'Go to resource' button is highlighted. The right sidebar contains sections for 'Cost Management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

Resource	Type	Status	Operation details
myVMMgmt	Microsoft.Compute/virtualMachines	OK	Operation details
myvmgmt709	Microsoft.Network/networkInterfa...	OK	Operation details
myVMMgmt-ip	Microsoft.Network/publicIpAddre...	OK	Operation details

Task 3: Associate each virtual machines network interface to it's application security group.

In this task, I was to associate each virtual machines network interface with the corresponding application security group. The myVMWeb virtual machine interface will be associated to the myAsgWebServers ASG. The myVMMgmt virtual machine interface will be associated to the myAsgMgmtServers ASG.

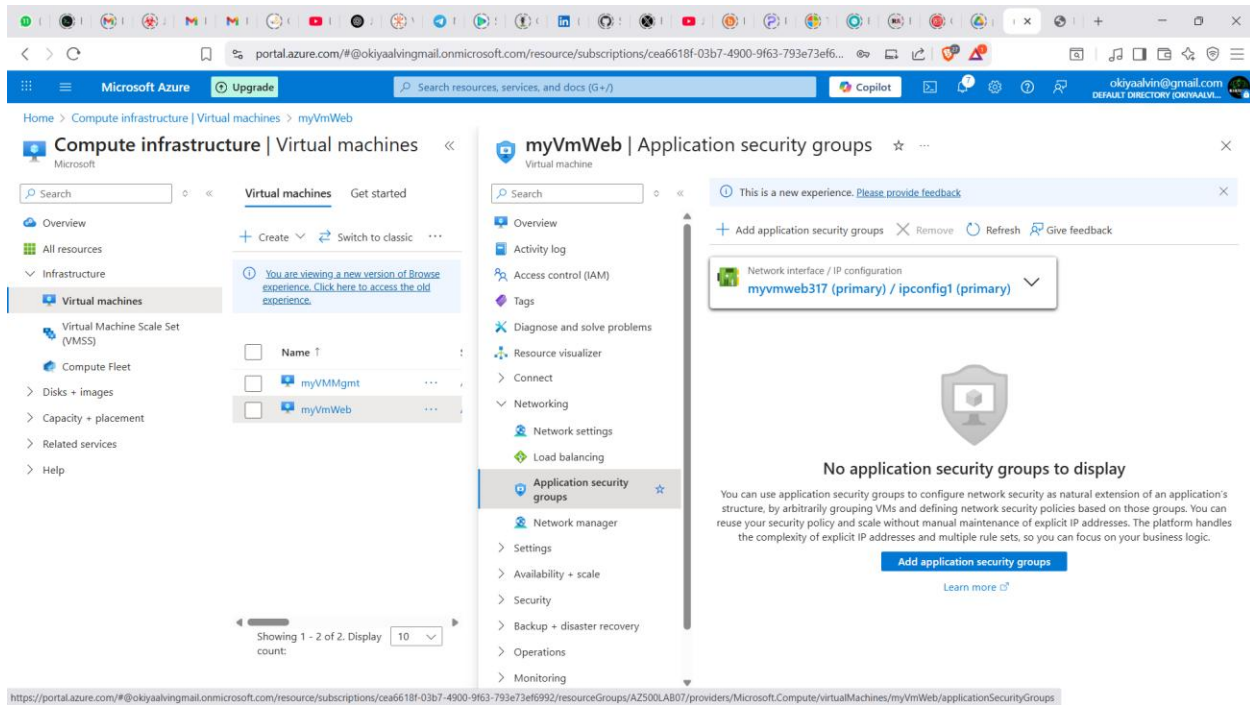
In the Azure portal, I navigated back to the **Virtual machines** blade and verified that both virtual machines are listed with the **Running** status. Then In the list of virtual machines, I clicked the **myVMWeb** entry



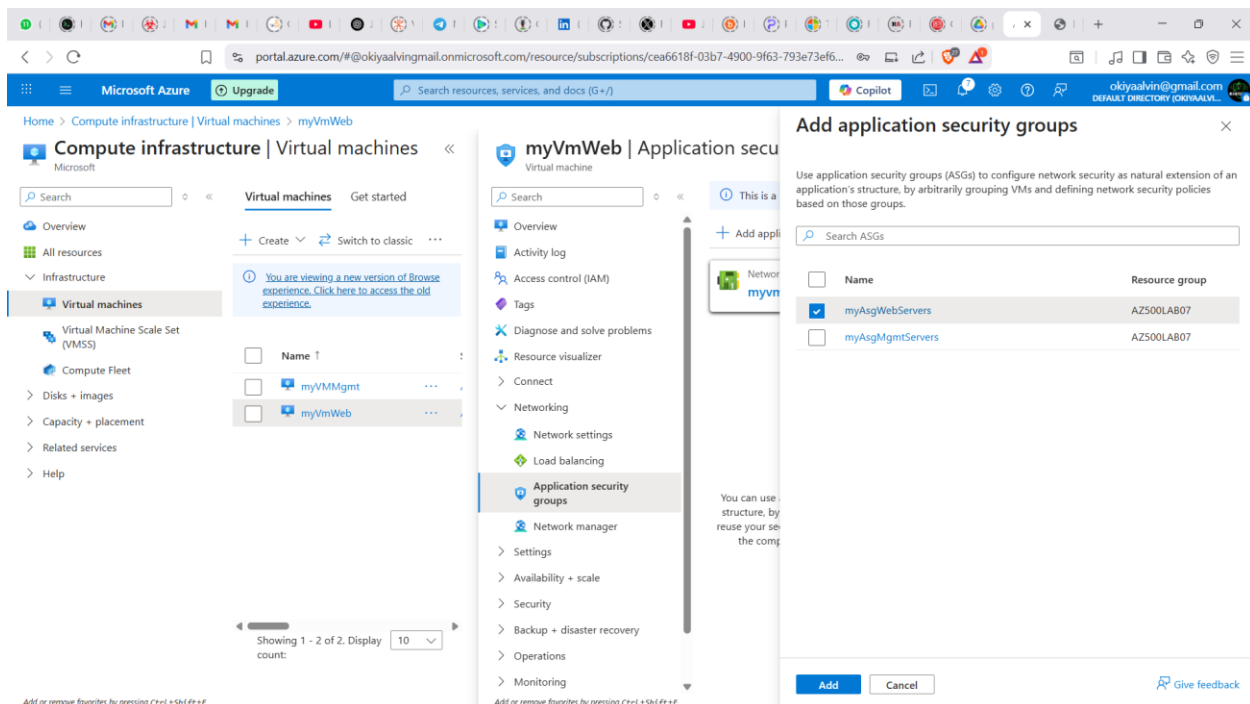
The screenshot shows the Azure portal interface for the 'Virtual machines' blade. The left sidebar contains navigation links for Overview, All resources, Infrastructure, Virtual machines, Virtual Machine Scale Set (VMSS), Compute Fleet, Disks + images, Capacity + placement, Related services, and Help. The main content area shows a list of virtual machines with columns for Name, Subscription, Resource Group, Location, Status, Operating system, Size, Public IP address, and Disks. Two virtual machines are listed: myVMMgmt and myVMWeb. Both are in the 'Running' status. The myVMWeb VM is selected, and its details are shown in the table below.

Name	Subscription	Resource Group	Location	Status	Operating system	Size	Public IP address	Disks
myVMMgmt	Azure subscript...	AZ500LAB07	East US	Running	Windows	Standard_D2s_v3	40.114.7.118	1
myVMWeb	Azure subscript...	AZ500LAB07	East US	Running	Windows	Standard_D2s_v3	172.203.139.180	1

On the **myVMWeb** blade, in the **Networking** section, I clicked **Network settings** and then, on the **myVMWeb | Networking settings** blade, I clicked the **Application security groups** tab.



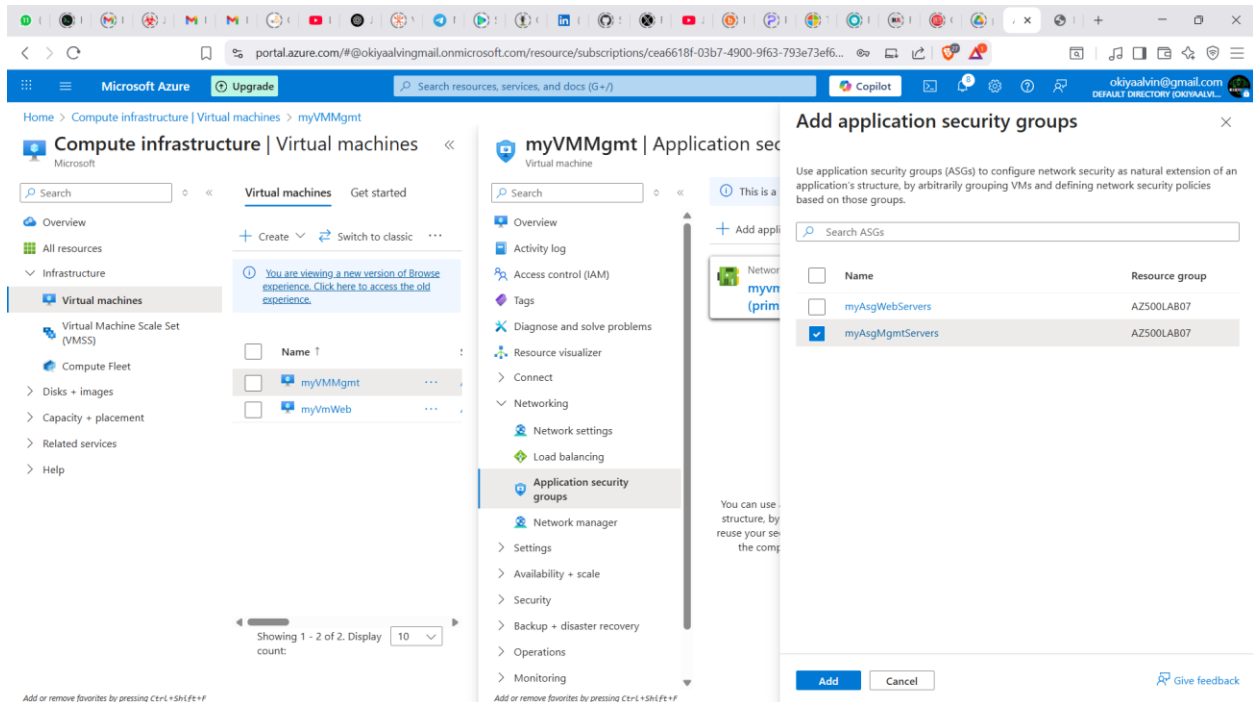
I Clicked + **Add application security groups**, then in the **Application security group** list, I selected **myAsgWebServers**, and then clicked **Add**.



I repeated the same process for the management server:

1. Navigated back to the **Virtual machines** blade and in the list of virtual machines, clicked the **myVMMgmt** entry.

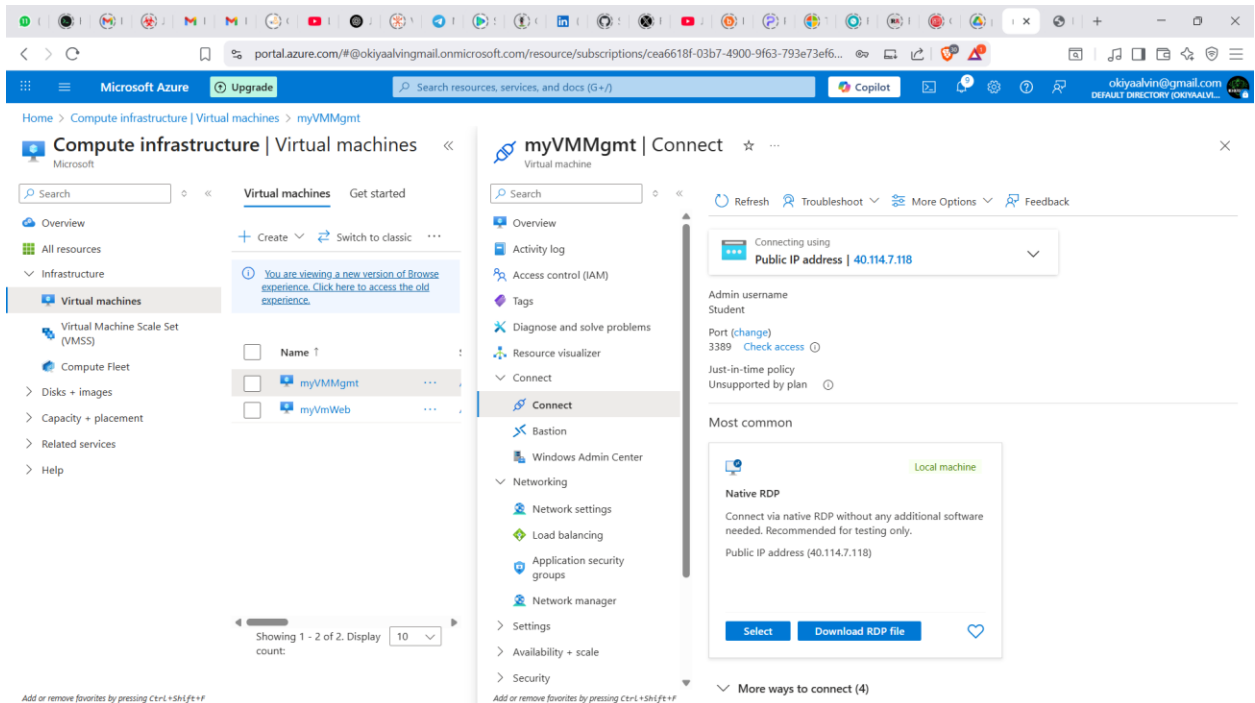
- On the **myVMMgmt** blade, in the **Networking** section, click **Networking settings** and then, on the **myVMMgmt | Networking settings** blade, click the **Application security groups** tab.
- Click **+ Add application security groups**, in the **Application security group** list, select **myAsgMgmtServers**, and then click **Save**.



Task 4: Test the network traffic filtering.

In this task, I was to test the network traffic filters. I should be able to RDP into the myVMMgmt virtual machine. I should be able to connect from the internet to the myVMWeb virtual machine and view the default IIS web page.

- I navigated back to the **myVMMgmt** virtual machine blade.
- On the **myVMMgmt** Overview blade, I clicked **Connect** and, in the drop down menu, I clicked **Connect**.



3. I downloaded the RDP file and used it to connect to the **myVMMgmt** Azure VM via Remote Desktop. When prompted to authenticate, I provided the credentials I had earlier created using username student:

However I encountered an error:

Remote Desktop Connection



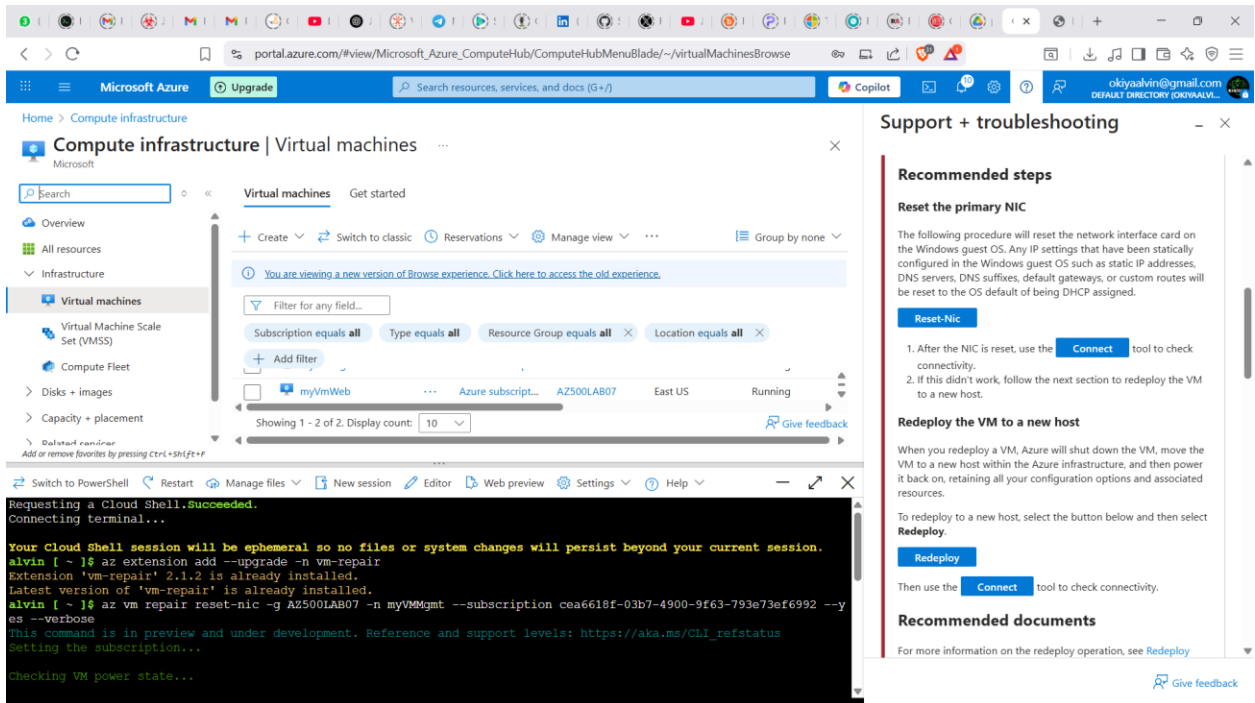
Remote Desktop can't connect to the remote computer for one of these reasons:

- 1) Remote access to the server is not enabled
- 2) The remote computer is turned off
- 3) The remote computer is not available on the network

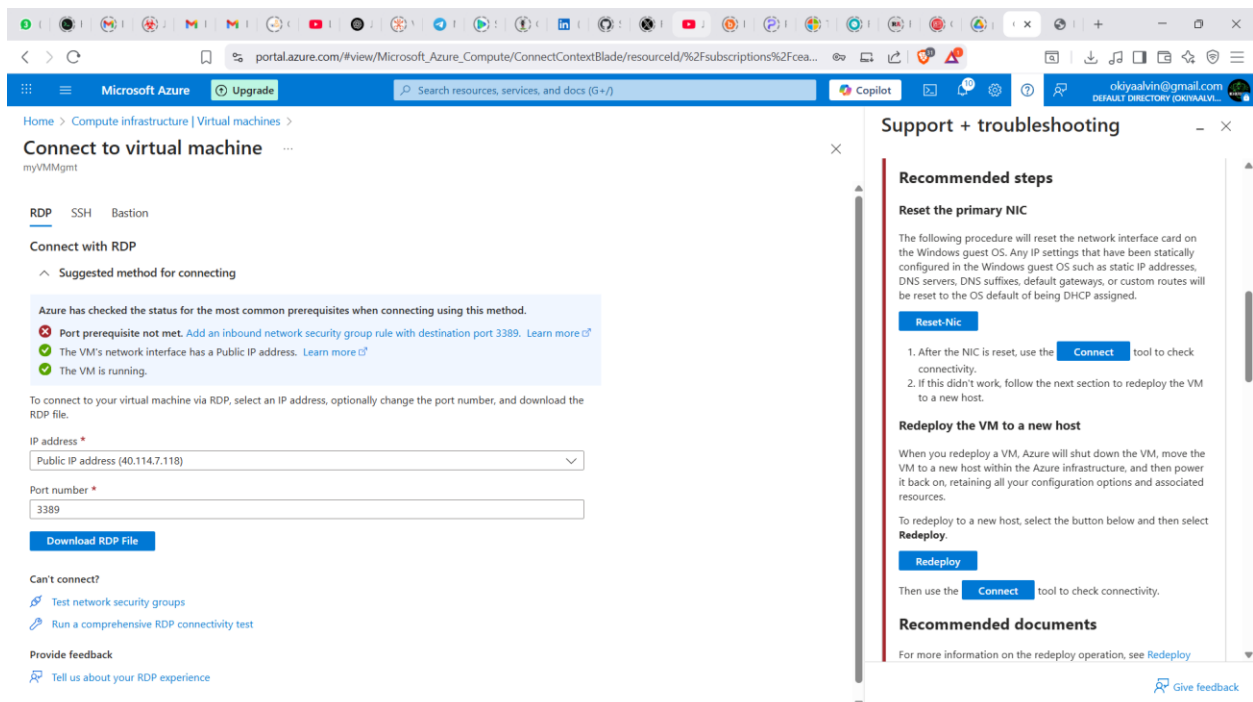
Make sure the remote computer is turned on and connected to the network, and that remote access is enabled.

 See details OK

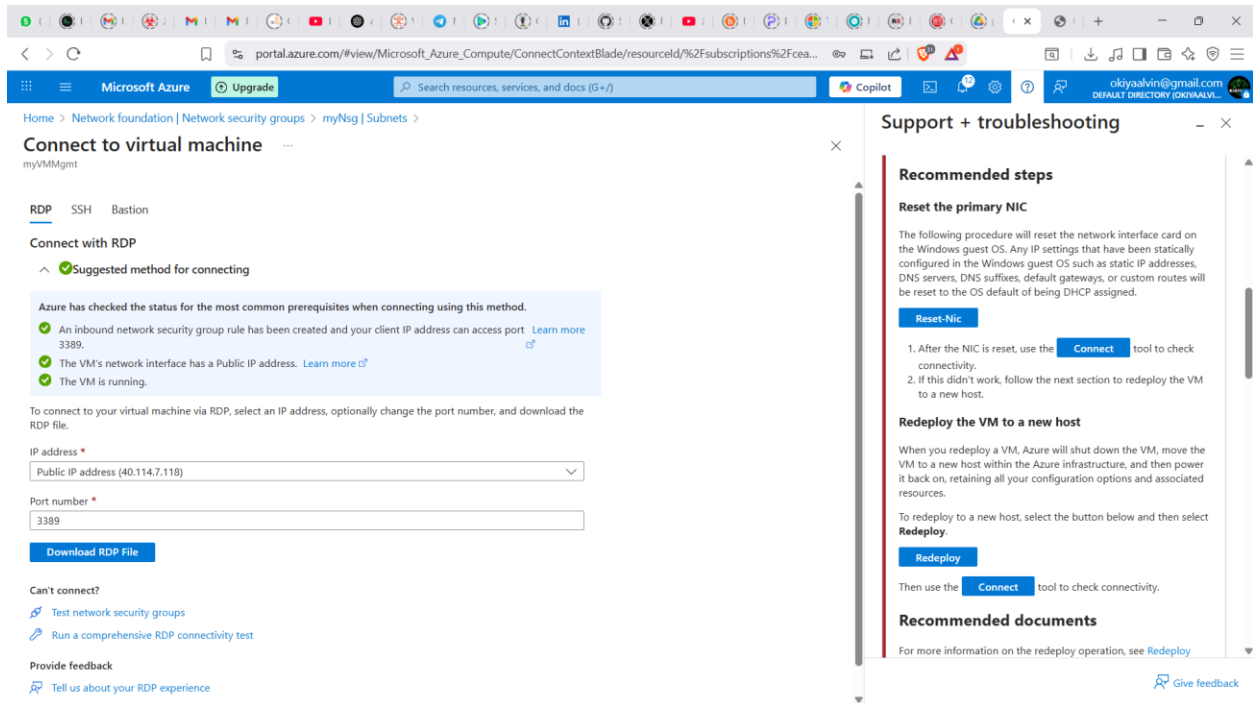
Started troubleshooting by resetting the NIC:



However it did not work, I used the connect tool to check the connectivity and I identified the issue below:



I then went to the Nsg subnets section to see if it had associated with the default subnet, and It hadn't, I repeated the process as highlighted in exercise 1 task 3, and it successfully connected



Microsoft Azure

portal.azure.com/#view/Microsoft_Azure_Compute/ConnectContextBlade/resourceId/%2Fsubscriptions%2Fcea...

Home > Network foundation | Network security groups > myNsg | Subnets >

Connect to virtual machine

myVMgmt

RDP SSH Bastion

Connect with RDP

^ Suggested method for connecting

Azure has checked the status for the most common prerequisites when connecting using this method.

- ✓ An inbound network security group rule has been created and your client IP address can access port 3389. [Learn more](#)
- ✓ The VM's network interface has a Public IP address. [Learn more](#)
- ✓ The VM is running.

To connect to your virtual machine via RDP, select an IP address, optionally change the port number, and download the RDP file.

IP address *

Public IP address (40.114.7.118)

Port number *

3389

Download RDP File

Can't connect?

- Test network security groups
- Run a comprehensive RDP connectivity test

Provide feedback

- Tell us about your RDP experience

Support + troubleshooting

Recommended steps

Reset the primary NIC

The following procedure will reset the network interface card on the Windows guest OS. Any IP settings that have been statically configured in the Windows guest OS such as static IP addresses, DNS servers, DNS suffixes, default gateways, or custom routes will be reset to the OS default of being DHCP assigned.

Reset-Nic

1. After the NIC is reset, use the **Connect** tool to check connectivity.
2. If this didn't work, follow the next section to redeploy the VM to a new host.

Redeploy the VM to a new host

When you redeploy a VM, Azure will shut down the VM, move the VM to a new host within the Azure infrastructure, and then power it back on, retaining all your configuration options and associated resources.

To redeploy to a new host, select the button below and then select **Redeploy**.

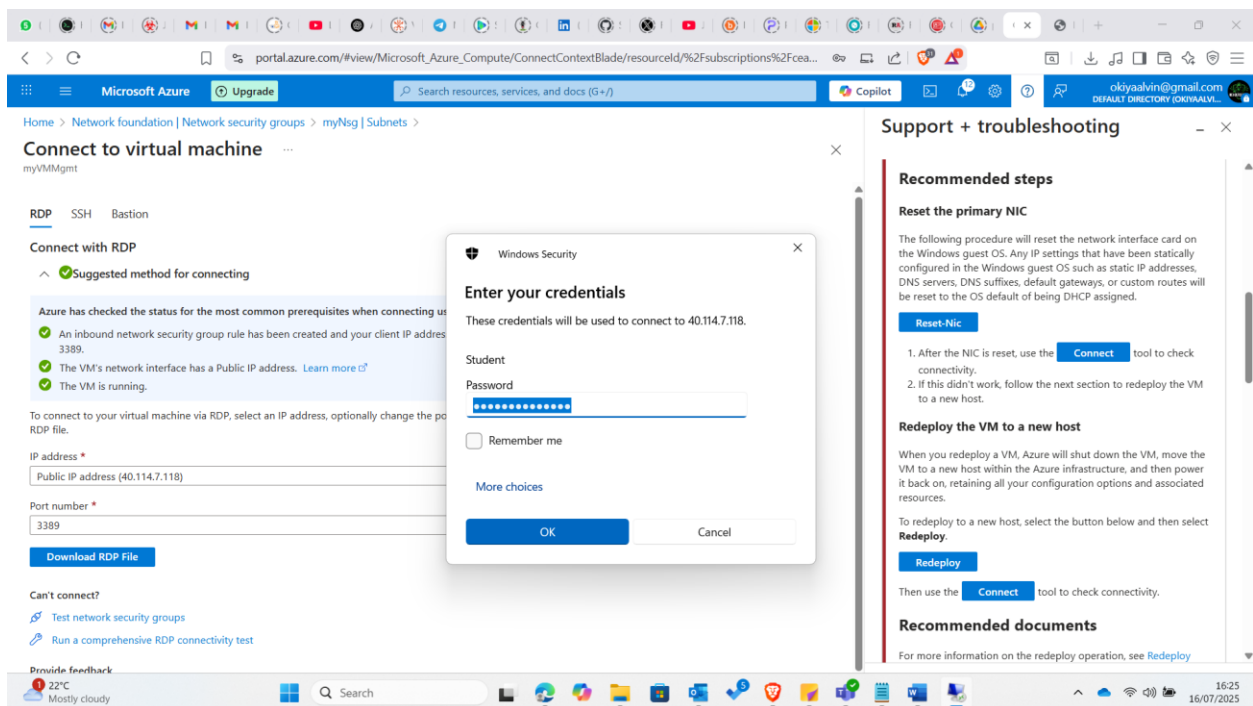
Redeploy

Then use the **Connect** tool to check connectivity.

Recommended documents

For more information on the redeploy operation, see [Redeploy](#)

Give feedback



Microsoft Azure

portal.azure.com/#view/Microsoft_Azure_Compute/ConnectContextBlade/resourceId/%2Fsubscriptions%2Fcea...

Home > Network foundation | Network security groups > myNsg | Subnets >

Connect to virtual machine

myVMgmt

RDP SSH Bastion

Connect with RDP

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IP address *

Public IP address (40.114.7.118)

Port number *

3389

Download RDP File

Can't connect?

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Redeploy

Then use the **Connect** tool to check connectivity.

Recommended documents

For more information on the redeploy operation, see [Redeploy](#)

Windows Security

Enter your credentials

These credentials will be used to connect to 40.114.7.118.

Student

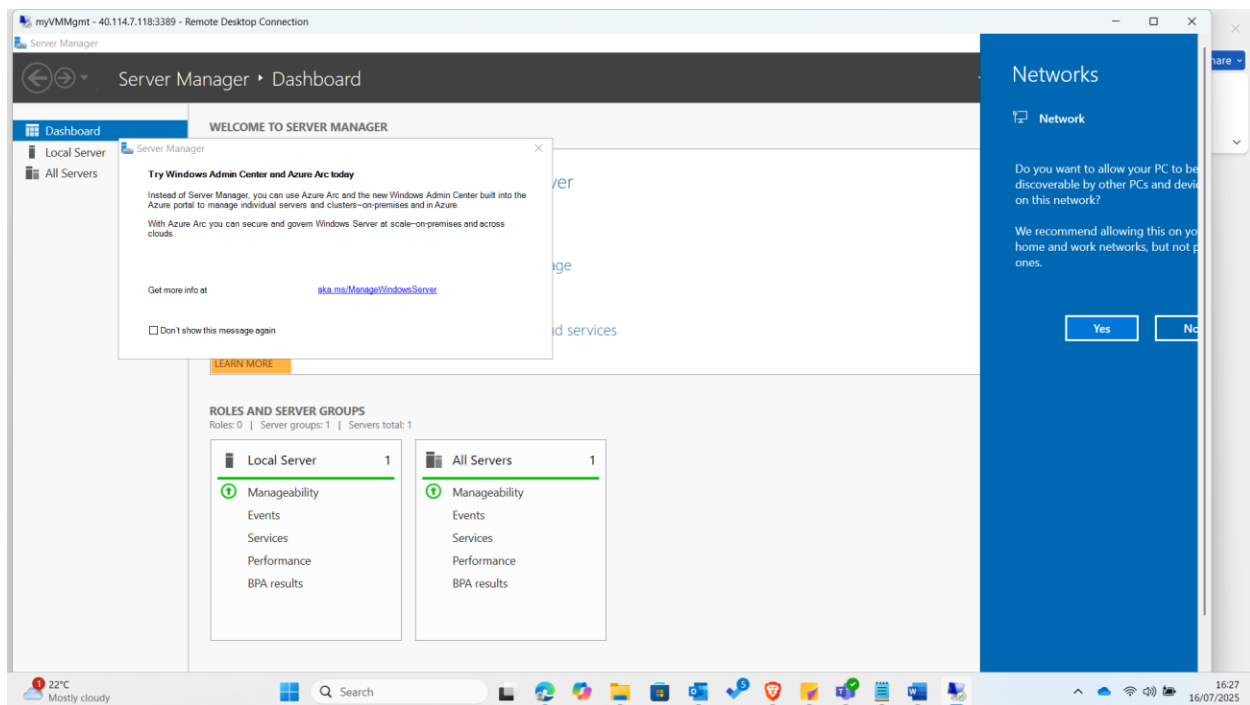
Password

Remember me

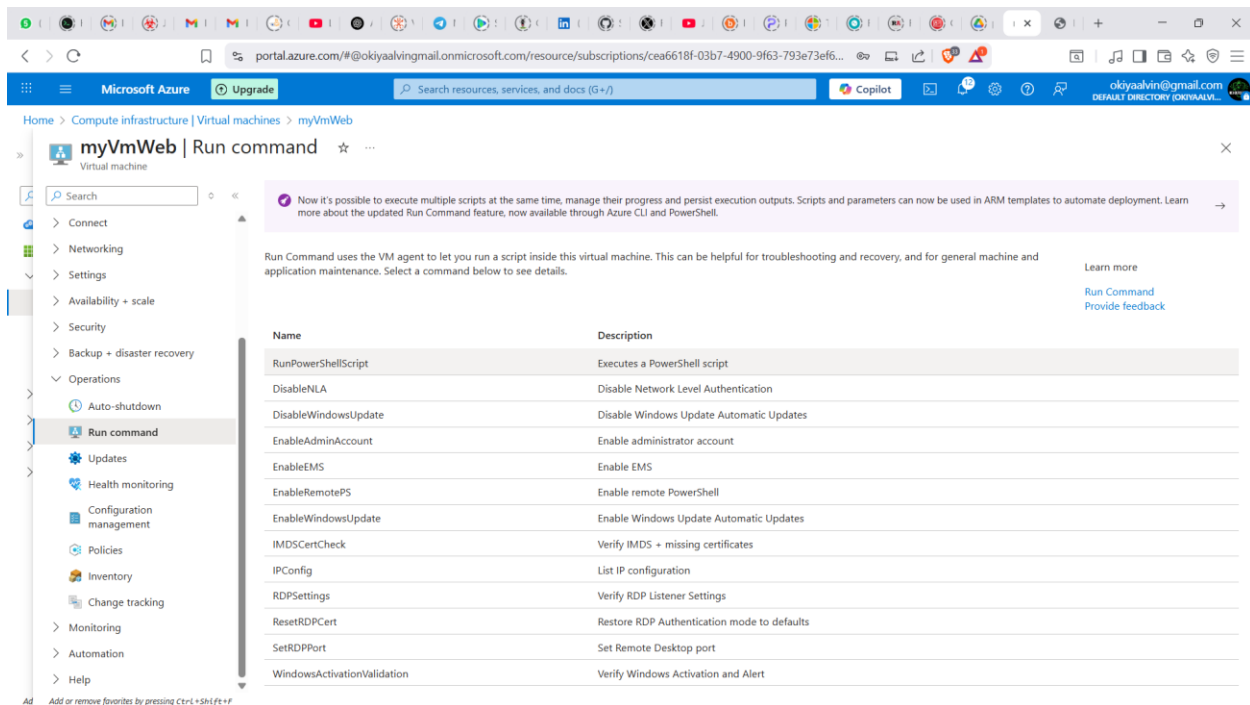
More choices

OK Cancel

Testing RDP connection:

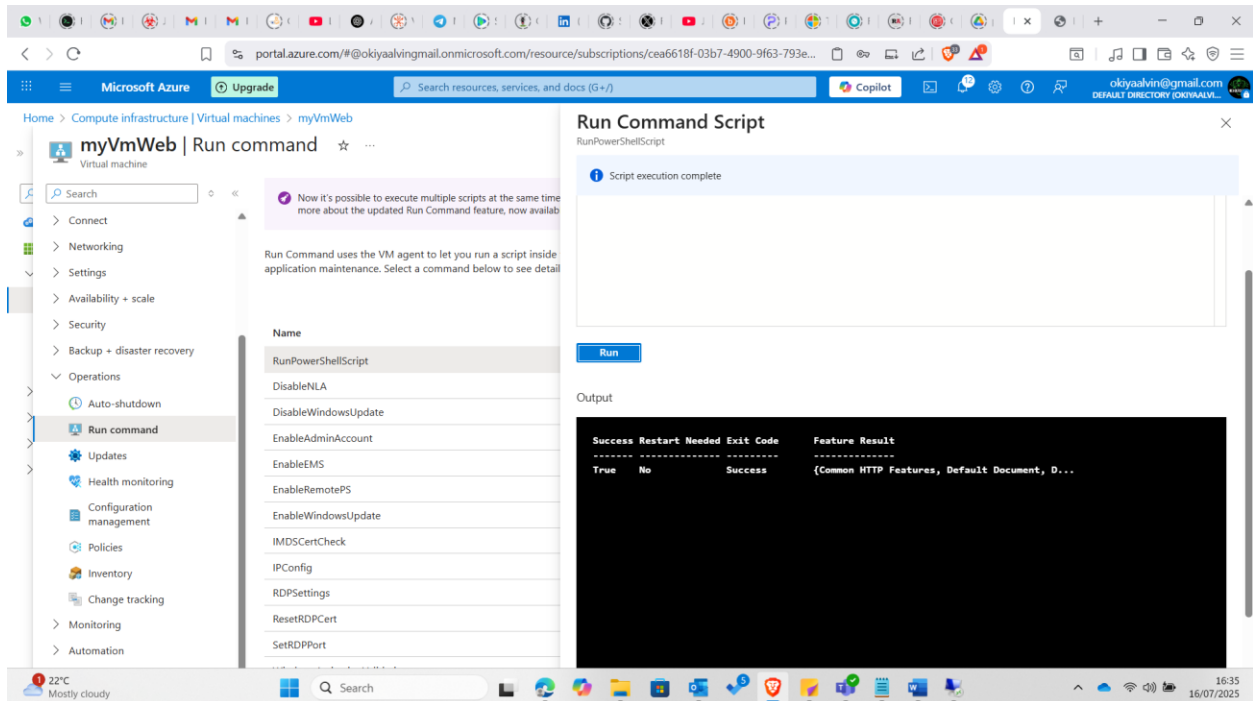


Next, In the Azure portal, I navigated to the **myVMWeb** virtual machine blade and On the **myVMWeb** blade, in the **Operations** section, I clicked **Run command** and then clicked **RunPowerShellScript**,

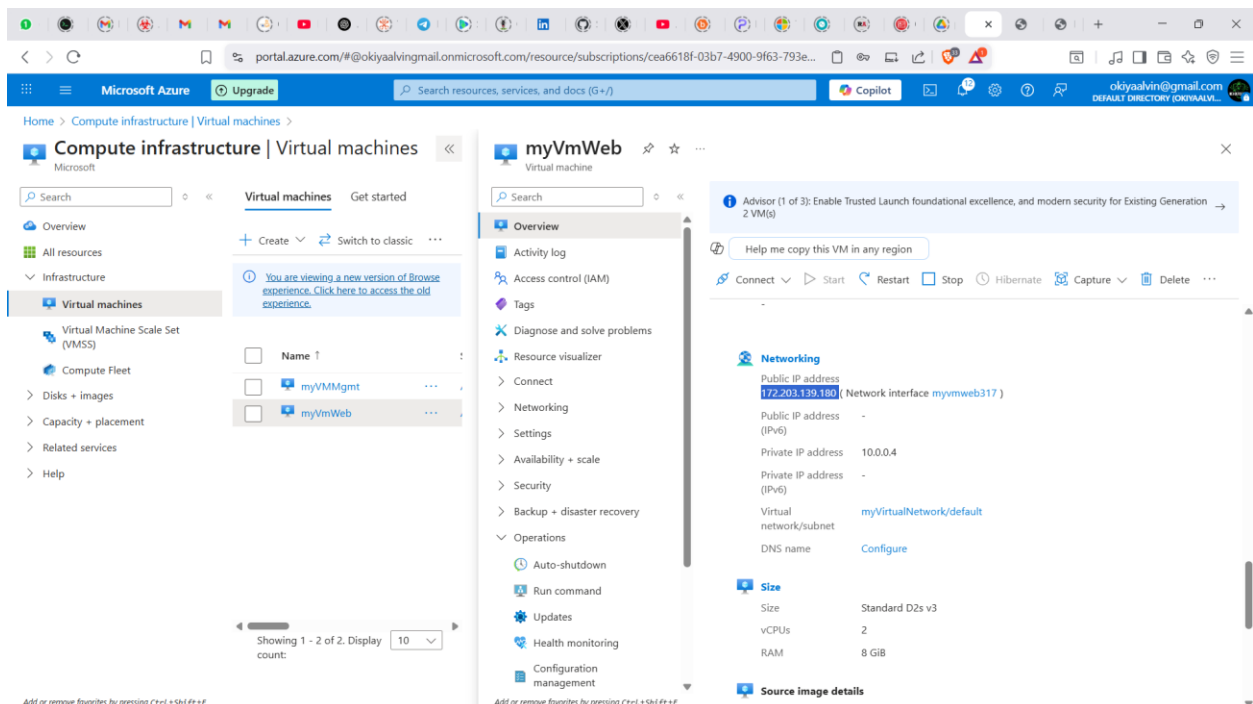


and ran the script below:

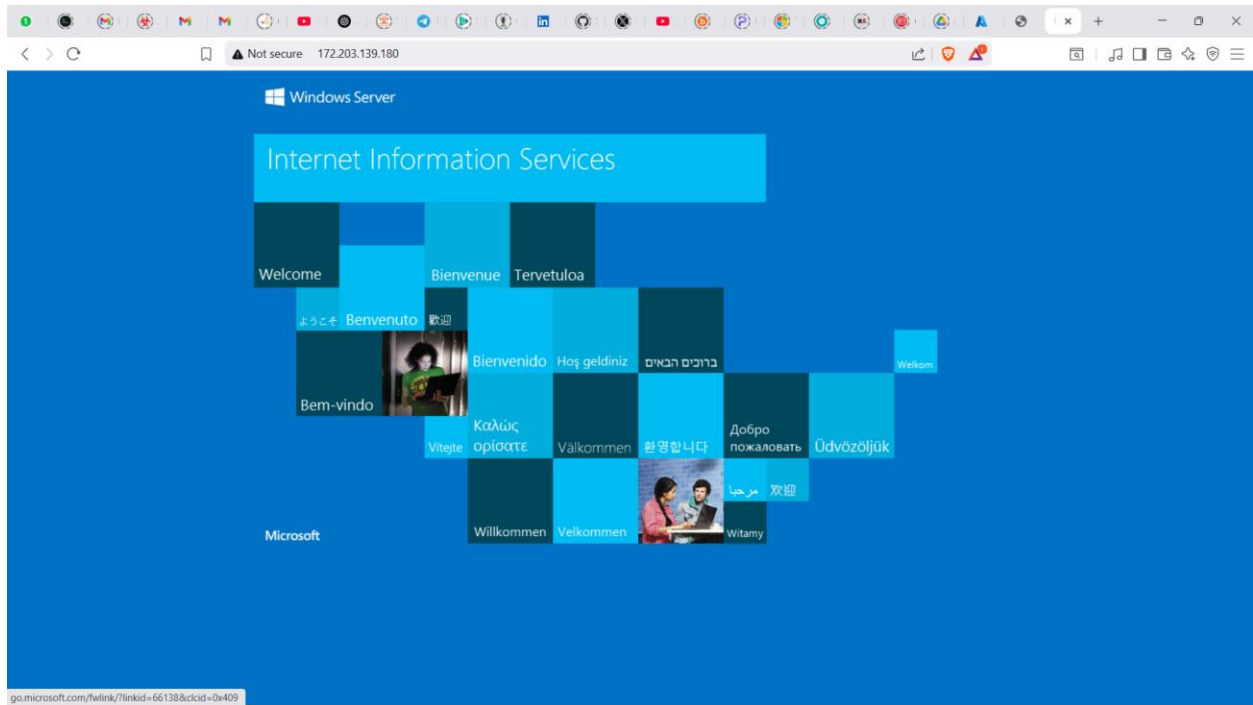
`Install-WindowsFeature -name Web-Server -IncludeManagementTools`



In the Azure portal, I navigated back to the **myVMWeb** blade. On the **myVMWeb** blade, I identified the **Public IP address** of the myVmWeb Azure VM



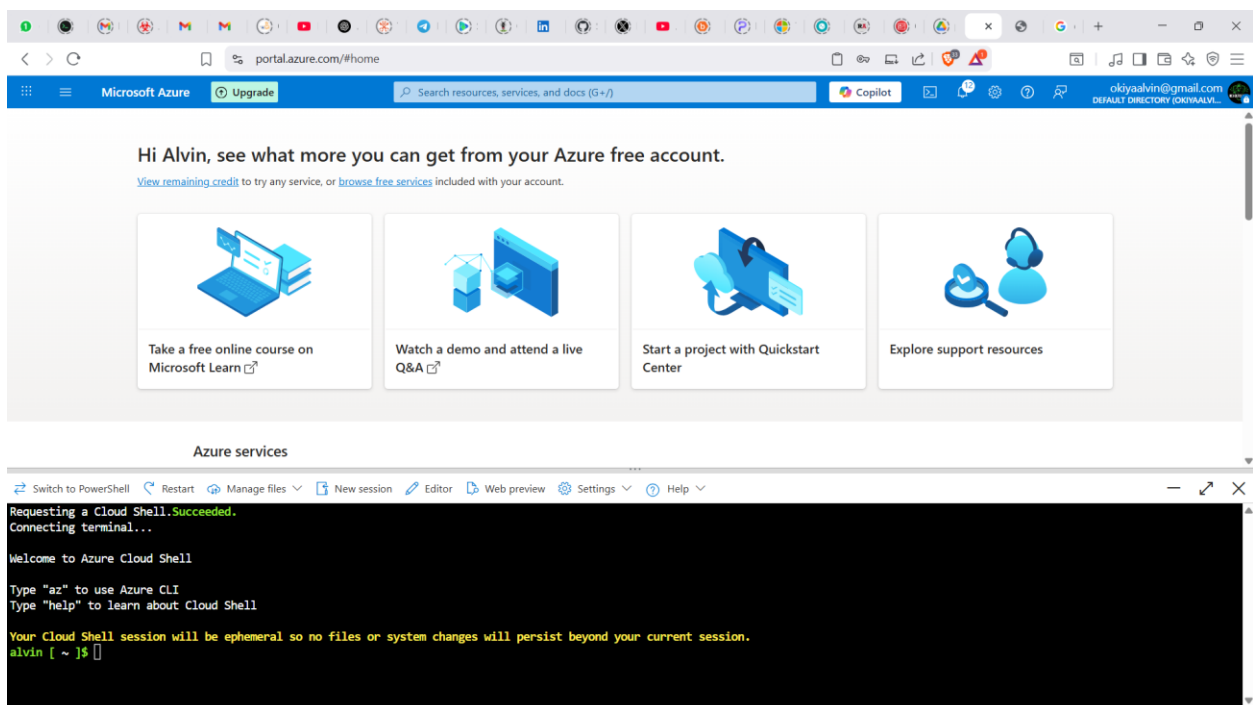
I opened a separate tab and on it, I used the ip address to access the web server:

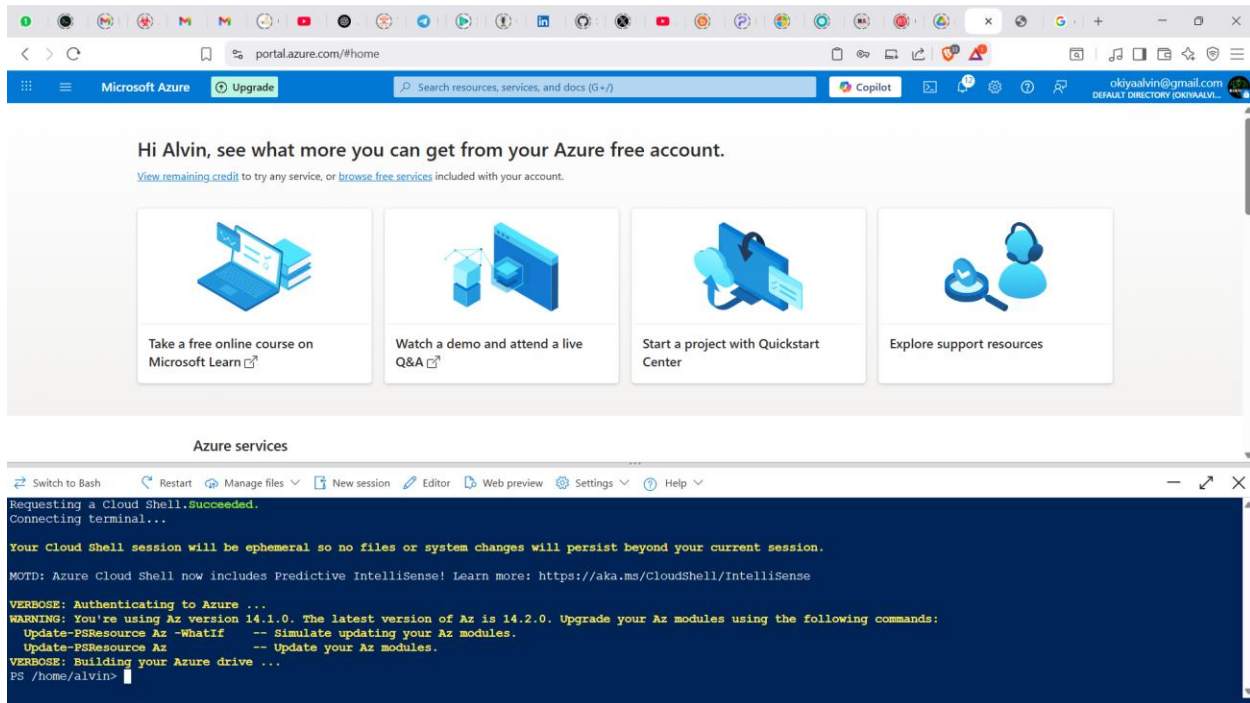


It displayed the default IIS welcome page confirming that the practical was successful.

Clean up resources

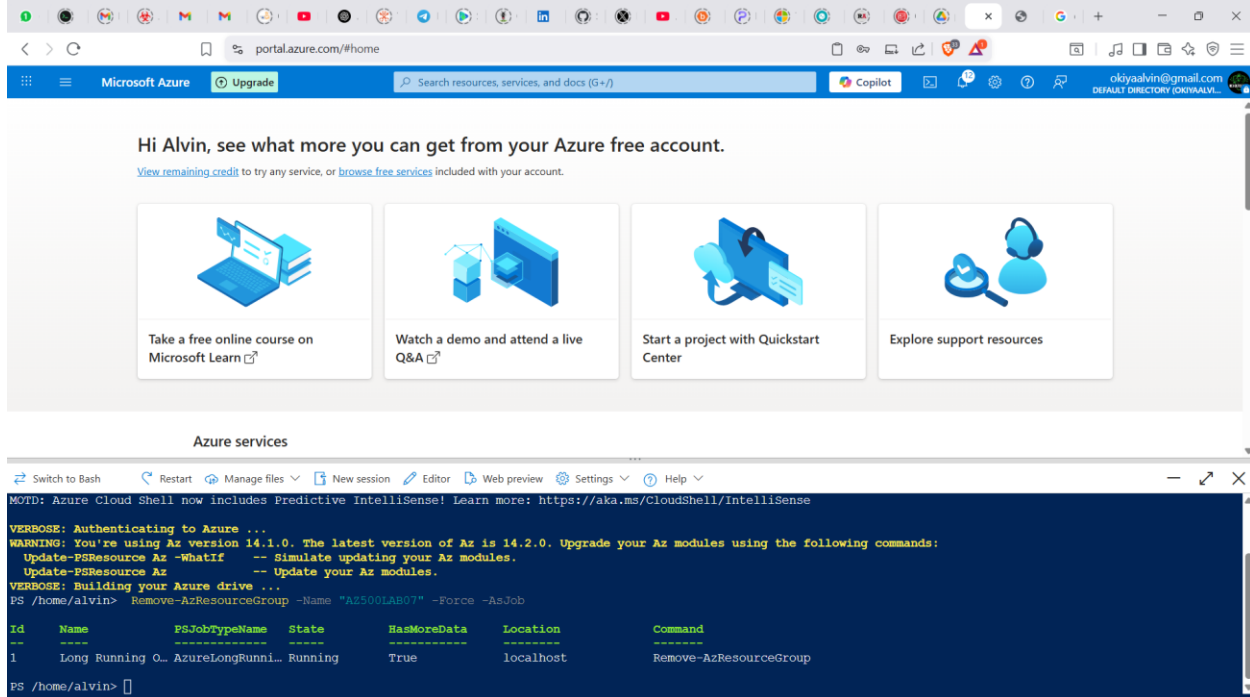
To prevent further consumption of the credits, I had to clean up resources. I opened the powershell on azure portal, it first brought a bash terminal but I switched to powershell:





I then ran the following command:

`Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob`



The clean up process started in the background.

Conclusion

Upon completing this lab, one will have successfully implemented a secure and functional virtual network infrastructure in Azure using NSGs and ASGs. The lab demonstrated how to isolate server groups, apply role-specific access control, and validate security configurations by testing RDP and web access. By associating virtual machines with the appropriate application security groups and applying targeted network security rules, this exercise reinforces core cloud security practices. These configurations not only protect Azure resources from unauthorized access but also ensure compliance with organizational security requirements. Finally, cleaning up unused resources prevents unnecessary costs and keeps the environment tidy.